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ORIGINAL PAPER

Intelligent multi-UAV path planning in complex environments: a hybrid optimization approach with direction-assisted search and adaptive refinement

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Abstract
Multi-UAV cooperative path planning in complex urban environments faces critical challenges in balancing exploration efficiency, convergence precision, and computational practicality. This study proposes a Hybrid Direction-Assisted Polar Light Optimization (HDPLO) algorithm that integrates three key innovations: directional heuristic mechanisms for intelligent search space expansion, adaptive crossover strategies for diversity maintenance, and CMA-ES integration for high-precision local refinement. Experimental validation using CEC2017 benchmarks and realistic multi-UAV scenarios (obstacle densities 7.71%-77.11%) demonstrates significant improvements: 38.9% better solution quality on benchmark functions, 45.7% faster convergence, 15.2% reduced path costs in UAV planning, and 11.1%-38.5% fewer turning maneuvers compared to state-of-the-art algorithms. Statistical analysis confirms performance reliability ($p<0.001$), while ablation studies verify each component's essential contribution. These findings establish HDPLO as a robust solution for intelligent UAV systems requiring high reliability and efficiency in real-world deployments.

Keywords Multi-UAV systems · Intelligent path planning · Hybrid optimization · Direction-assisted search · Adaptive refinement · Complex environments

List of Abbreviations

Abbreviation	Full Form
UAV	Unmanned Aerial Vehicle
HDPLO	Hybrid Direction-Assisted Polar Light Optimization
PLO	Polar Light Optimization
CMA-ES	Covariance Matrix Adaptation Evolution Strategy
PSO	Particle Swarm Optimization
GA	Genetic Algorithm
DE	Differential Evolution
ACO	Ant Colony Optimization
RRT	Rapidly-exploring Random Tree
PRM	Probabilistic Roadmap
GWO	Grey Wolf Optimizer
MGO	Multi-objective Grey Wolf Optimizer
SAO	Snow Ablation Optimizer
BKA	Black-winged Kite Algorithm
ALCPSO	PSO with Aging Leader and Challengers
OBSCA	Opposition-Based Sine Cosine Algorithm
CEC	Congress on Evolutionary Computation

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1 Introduction

Unmanned Aerial Vehicle (UAV) technology has experienced unprecedented rapid development over the past decade, expanding from military reconnaissance to civilian sectors including environmental monitoring, logistics delivery, agricultural plant protection, and emergency rescue (Liu and Zheng 2023; Alexan et al. 2024; Vasiloni et al. 2023). With smart city construction advancing and airspace management policies opening, multi-UAV cooperative systems have become essential for large-scale, high-efficiency tasks

(Murray and Chu 2015; Ha et al. 2018; Phung et al. 2017b; Betalo et al. 2025b).

Research gap and motivation Despite extensive research on UAV path planning, existing methods exhibit critical limitations in practical scenarios. Traditional graph-based algorithms (A*, Dijkstra) face exponential complexity growth in dynamic environments (Hart et al. 1968; Stentz 1994; Dijkstra 1959). Sampling-based methods (RRT, PRM) suffer from slow convergence and unstable solution quality in high-dimensional spaces (LaValle 1998; Kavraki et al. 1996). Recent meta-heuristic algorithms show promise but commonly experience premature convergence, limited search space exploration, and insufficient local refinement (Kim et al. 2024; Russell and Norvig 2020). Multi-UAV coordination amplifies these challenges by requiring simultaneous consideration of individual path optimization, collision avoidance, communication delays, and multiple complex constraints (Choset et al. 2005; Kennedy and Eberhart 1995).

Proposed solution To address these challenges, this paper proposes a **Hybrid Direction-Assisted Polar Light Optimization algorithm (HDPLO)** that achieves performance breakthroughs through innovative integration of directional heuristic mechanisms, improved crossover strategies, and covariance matrix adaptation evolution strategies. Building upon the Polar Light Optimization (PLO) framework (Yuan et al. 2024), HDPLO forms a hybrid optimization scheme that overcomes PLO's limitations in local refinement and population diversity maintenance.

Main contributions Specifically, the contributions of this paper include:

(1) Design of a directional heuristic search mechanism that expands traditional 8-neighborhood constraints through geometric relationship analysis, improving global search efficiency by approximately 300% coverage expansion in complex obstacle environments (Chen et al. 2012; Holland 1992);
(2) Development of a horizontal-vertical bidirectional crossover strategy maintaining population diversity through multi-level information exchange, achieving 22.3% improvement in solution quality while preventing premature convergence (Dorigo et al. 2006; Storn and Price 1997);
(3) Innovative CMA-ES adaptive integration providing high-precision local search with parameter self-adjustment, contributing 31.2% performance enhancement and significantly improving convergence accuracy (Abd Elaziz et al. 2017; Mirjalili et al. 2014);
(4) Construction of a comprehensive multi-constraint dynamic path planning model incorporating flight safety, path efficiency, communication constraints, and practical UAV operational requirements (Wang et al. 2024; Yuan et al. 2024).

Paper organization Section 2 reviews existing path planning algorithms and identifies their limitations; Section 3 details the HDPLO algorithm design; Section 4 presents multi-level experimental validation; Section 5 discusses performance characteristics and application potential; Section 6 concludes with future research directions (Peraza-Vázquez et al. 2024).

2 Related work

2.1 Traditional path planning algorithms

Traditional path planning algorithms are mainly divided into two major categories: graph search-based and sampling-based methods, which have laid important theoretical foundations in the path planning field. The A* algorithm proposed by Hart et al. (1968); Li et al. (2024) guides the search process through heuristic functions and can guarantee finding optimal paths in static environments, but its computational complexity grows exponentially with environmental complexity. The Dynamic A* algorithm (D*) (Stentz 1994), although supporting incremental path updates in dynamic environments, has limited efficiency when dealing with frequently changing dynamic obstacles. The Dijkstra algorithm, as a classic shortest path algorithm, can guarantee solution optimality but requires traversing the entire search space, resulting in enormous computational overhead in large-scale environments (Dijkstra 1959).

Sampling-based path planning methods construct paths through random sampling, possessing certain flexibility and adaptability. The Rapidly-exploring Random Tree (RRT) algorithm (LaValle 1998) implements path search through random sampling and tree expansion, but has slow convergence speed in high-dimensional spaces and difficulty guaranteeing path optimality. The Probabilistic Roadmap (PRM) method (Kavraki et al. 1996) preprocesses environmental information by constructing roadmaps, but requires frequent roadmap reconstruction in dynamic environments, resulting in low computational efficiency. The Voronoi diagram method (Kim et al. 2024), although capable of generating safe paths away from obstacles, has high construction and maintenance costs in complex environments.

Although these traditional methods perform well under specific conditions, they generally have several key limitations: first, computational complexity grows rapidly with envi-

ronmental complexity, making it difficult to meet real-time application requirements; second, most methods assume static environments and have limited capability for handling dynamic obstacles; finally, these methods typically consider only single optimization objectives and have difficulty handling multi-constraint optimization problems in multi-UAV coordination (Russell and Norvig 2020; Choset et al. 2005; Huo et al. 2025).

2.2 Meta-heuristic optimization algorithms

To overcome the limitations of traditional methods, researchers have begun applying meta-heuristic optimization algorithms to path planning problems. The Particle Swarm Optimization (PSO) algorithm (Kennedy and Eberhart 1995) has been widely applied in path planning due to its simple implementation and fast convergence speed, but it easily falls into local optima and has limited performance in complex multimodal problems. The PSO algorithm with aging leader and challengers (ALCPSO) proposed by Chen et al. (2012) alleviates premature convergence problems through dynamic leader particle updates, but still has issues with low search efficiency in high-dimensional complex environments.

Genetic Algorithm (GA) (Holland 1992) achieves global optimization by simulating natural evolution processes and demonstrates good global search capability in path planning, but has relatively slow convergence speed and parameter settings significantly affect performance. Ant Colony Optimization (ACO) algorithm (Dorigo et al. 2006) implements positive feedback search through pheromone mechanisms and is particularly suitable for discrete path planning problems, but its application in continuous space problems is limited. Differential Evolution (DE) algorithm (Storn and Price 1997) performs excellently in numerical optimization due to its simple and effective mutation strategy, but requires complex constraint handling mechanisms when dealing with constrained optimization problems.

In recent years, researchers have also proposed a series of improved meta-heuristic algorithms. The Opposition-Based Sine Cosine Algorithm (OBSCA) developed by Abd Elaziz et al. (2017) enhances the algorithm's exploration capability through opposition learning strategies, but still has deficiencies in local search precision. Grey Wolf Optimizer (GWO) and its variants (Mirjalili et al. 2014) achieve optimization search by simulating wolf pack hunting behavior and perform excellently on certain problems, but lack effective diversity maintenance mechanisms.

2.3 Emerging optimization algorithms and path planning applications

With the development of bionics and intelligent computing theory, many new optimization algorithms have emerged in recent years. The Black-winged Kite Algorithm (BKA) proposed by Wang et al. (2024) simulates the attack and migration behaviors of black-winged kites and performs excellently in handling high-dimensional optimization problems, but has issues such as slow convergence speed and insufficient local search capability in complex constrained environments. The Polar Light Optimizer (PLO) developed by Yuan et al. (2024); Zhong et al. (2025) achieves optimization search by simulating the dynamic change process of aurora phenomena and has certain advantages in global exploration, but still has room for improvement in local refinement and population diversity maintenance.

Some researchers have attempted to apply emerging algorithms to specific path planning problems. The Horned Lizard Optimization Algorithm (HLOA) proposed by Peraza-Vázquez et al. (2024) demonstrates unique advantages in certain engineering optimization problems, but lacks targeted design for complex constraint handling in multi-UAV cooperative path planning. The Polar Light Optimizer (PLO) developed by Yuan et al. (2024) achieves optimization search by simulating the dynamic change process of aurora phenomena and has certain advantages in global exploration, but still has room for improvement in local refinement and population diversity maintenance. This paper is precisely based on the PLO algorithm framework, integrating multiple optimization strategies to compensate for its deficiencies and enhance its performance.

It is worth noting that Covariance Matrix Adaptation Evolution Strategy (CMA-ES) (Hansen et al. 2003), as an efficient local search algorithm, performs excellently in continuous optimization problems, but has limitations when dealing with complex constraints and multimodal problems as an independent algorithm. Some researchers have attempted to combine CMA-ES with other algorithms, but most work lacks systematic theoretical analysis and targeted design optimization (Auger and Hansen 2005; Loshchilov et al. 2012).

2.4 Special challenges in multi-UAV cooperative path planning

Multi-UAV cooperative path planning faces more challenges compared to single-UAV path planning. Phung et al. (2017a) emphasized the complexity of multi-constraint optimization in UAV vision-based inspection path planning, including obstacle avoidance constraints, flight dynamics constraints, and mission coverage constraints. Betalo et al. (2025a) pointed out in their research on multi-UAV charging and path planning that multi-UAV systems also need to consider communication constraints, coordination constraints, and resource competition issues.

Existing multi-UAV path planning methods are mainly divided into centralized and distributed categories. Although

centralized methods can obtain globally optimal solutions, computational complexity grows exponentially with the number of UAVs, resulting in poor scalability (Otte and Frazzoli 2016). Although distributed methods have good scalability, they have difficulty guaranteeing global optimality and may produce deadlock problems when handling multi-UAV conflicts (Van Den Berg et al. 2011).

Table 1 Comparative analysis of path planning algorithms

Category	Algorithm	Key Strengths	Primary Limitations	Multi-UAV Suitability
Traditional Graph-Based	A* (Hart et al. 1968; Li et al. 2024)	Optimal in static env.; Completeness guarantee	Exponential complexity growth; Memory intensive	Low (poor scalability)
	D* (Stentz 1994)	Incremental replanning; Dynamic support	Limited efficiency with frequent changes	Low (high replanning overhead)
	Dijkstra (1959)	Guaranteed optimality; Simplicity	Explores entire space; Slow in large envs	Low (computational burden)
	Kim et al. (2024)	Safe paths away from obstacles	High construction/maintenance cost	Low (complex in 3D)
Sampling-Based	RRT (LaValle 1998)	Probabilistic completeness; High-dim capability	Slow convergence; No optimality guarantee	Medium (needs coordination layer)
	RRT* (Otte and Frazzoli 2016)	Asymptotic optimality; Flexibility	Computationally expensive; Slow	Medium (better with pruning)
	PRM (Kavraki et al. 1996)	Effective preprocessing; Reusable roadmaps	Requires frequent reconstruction in dynamic envs	Medium (shared roadmap possible)
Classic Meta-heuristic	PSO (Kennedy and Eberhart 1995)	Fast convergence; Simple implementation	Premature convergence; Local optima trapping	Medium (diversity issues)
	ALCPSO (Chen et al. 2012)	Improved diversity vs PSO; Dynamic leaders	Still limited in complex multimodal problems	Medium (better than PSO)
	GA (Holland 1992)	Strong global search; Population-based	Slow convergence; Parameter sensitive	Medium (complex tuning needed)
	ACO (Dorigo et al. 2006)	Distributed search; Positive feedback	Slow in continuous space; Pheromone tuning	Medium (good for discrete)
	DE (Storn and Price 1997)	Effective mutation strategy; Numerical strength	Complex constraint handling; Limited diversity	Medium (needs enhancement)
	GWO (Mirjalili et al. 2014)	Good exploitation; Bio-inspired	Lacks diversity maintenance; Premature convergence	Medium (needs improvement)
Emerging Algorithms	PLO (Yuan et al. 2024; Zhong et al. 2025)	Novel exploration mechanism; Global search	Limited local refinement; Insufficient diversity	Medium (foundation for HDPLO)
	BKA (Wang et al. 2024)	High-dimensional capability; Nature-inspired	Slow convergence; Weak local search	Low (needs significant improvement)
	SAO (Deng and Liu 2023)	Physical phenomenon-based; Exploration	Convergence precision issues	Medium (moderate performance)
	CMA-ES (Hansen et al. 2003)	Excellent local search; Adaptive covariance	Limited global exploration; Local optima risk	Low (needs global component)
Hybrid/Learning	DRL-based (Liu et al. 2022; Bayerlein et al. 2021)	Adaptive learning; Real-time inference	Requires extensive training; Generalization issues	Medium-High (after training)
	Hybrid approaches (Xie et al. 2023; Das et al. 2016)	Combines strengths; Balanced performance	Implementation complexity; Tuning required	High (domain-specific)
Proposed	**HDPLO**	**Balanced exploration-exploitation; High precision; Multi-component synergy; Strong robustness**	**Moderate computational overhead; Requires parameter configuration**	**High (specifically designed)**

To provide a systematic overview of the path planning landscape, Table 1 summarizes the representative algorithms discussed in this section, comparing their key characteristics, advantages, limitations, and suitability for multi-UAV cooperative scenarios. This comparison highlights the motivation for developing HDPLO as a hybrid approach that addresses the collective limitations of existing methods.

Comprehensive analysis of existing research reveals that although there has been extensive work on path planning algorithms, existing methods still have the following key deficiencies in the specific scenario of multi-UAV cooperative path planning: (1) traditional algorithms have high computational complexity and difficulty handling large-scale complex environments; (2) existing meta-heuristic algorithms commonly suffer from slow convergence speed and tendency to fall into local optima; (3) lack of effective mechanisms to balance global exploration and local exploitation; (4) insufficient capability for handling complex constraints in multi-UAV cooperation; (5) algorithm robustness and adaptability need improvement (Goerzen et al. 2010; Aggarwal and Kumar 2020; Zhang et al. 2020).

Table 1 reveals a clear trend: traditional methods offer theoretical guarantees but lack scalability, classic metaheuristics provide good exploration but suffer from premature convergence or slow refinement, and emerging single-strategy algorithms address specific weaknesses but lack comprehensive performance. This gap motivates HDPLO's design philosophy: systematically integrating directional heuristics (for intelligent exploration), adaptive crossover (for diversity maintenance), and CMA-ES (for precision refinement) to achieve balanced, robust performance specifically tailored for multi-UAV scenarios.

Therefore, developing an efficient multi-UAV path planning algorithm that can effectively combine global search and local refinement, possess strong constraint handling capabilities, and be suitable for complex dynamic environments has important theoretical significance and practical value. The HDPLO algorithm proposed in this paper, based on PLO with direction assistance and multi-strategy hybrid optimization, is precisely an innovative solution designed to address these challenges.

3 Methods

3.1 Algorithm design philosophy and problem statement

Table 2 provides a comprehensive list of all mathematical symbols and notations used throughout this paper, including their descriptions and default values where applicable.

Multi-UAV dynamic path planning presents a complex multi-objective constrained optimization challenge that requires balancing multiple competing objectives while satisfying safety and operational constraints (Phung et al. 2017a; Betalo et al. 2025a). Existing meta-heuristic algorithms face three fundamental limitations when applied to this domain: (1) restricted search space exploration in grid-based environments (Stentz 1994; Dijkstra 1959), (2) premature convergence due to insufficient population diversity maintenance (Kennedy and Eberhart 1995; Chen et al. 2012), and (3) inadequate local refinement capabilities for high-precision constraint satisfaction (Hansen et al. 2003; Auger and Hansen 2005).

To address these limitations, we propose the Hybrid Direction-Assisted Polar Light Optimization (HDPLO) algorithm, which enhances the basic Polar Light Optimization (PLO) framework (Yuan et al. 2024) through three targeted innovations:

- **Directional Heuristic Mechanism**: Expands search space beyond traditional 8-neighborhood constraints

Table 2 Notation and symbols

Symbol	Description
X_i	Path sequence of the i-th UAV: $[x_1, y_1, z_1, ..., x_n, y_n, z_n]$
n	Number of waypoints in a path
N_{UAV}	Total number of UAVs
m	Number of obstacles in the environment
$d_{j,k}$	Euclidean distance from waypoint j to obstacle k (meters)
R_k	Radius of obstacle k (meters)
α	Threat intensity coefficient (default: 1000)
β	Decay index for obstacle threat (default: 1.5)
ϵ	Small constant for numerical stability (default: 0.1)
ω_j	Weight for altitude deviation at waypoint j
$z_{optimal}$	Optimal flight altitude (meters)
θ_j	Heading angle at waypoint j (radians)
ϕ_j	Pitch angle at waypoint j (radians)
d_{comm}	Maximum communication distance (meters)
$H(\cdot)$	Heaviside step function: $H(x) = \begin{cases} 1 & x > 0 \\ 0 & x \leq 0 \end{cases}$
W_j	Wind speed at waypoint j (m/s)
h_{wind}	Wind decay height parameter (meters)
δ	Small constant preventing division by zero
N	Population size
T	Maximum number of iterations
D	Problem dimension
K	CMA-ES activation interval (generations)
P_c	Crossover probability
μ	Number of elite individuals for CMA-ES
σ	Step size in CMA-ES
C	Covariance matrix in CMA-ES

- **Improved Crossover Strategy**: Maintains population diversity through horizontal-vertical information exchange
- **CMA-ES Integration**: Provides high-precision local search and parameter adaptation

The synergistic combination of these components enables dynamic balance between global exploration and local exploitation, addressing the core challenges of multi-UAV path planning in complex environments.

3.2 Multi-objective problem formulation

The multi-UAV path planning problem is mathematically formulated as a constrained optimization problem where each UAV must find an optimal trajectory from start to goal position while satisfying multiple objectives, following the multi-objective optimization framework established in evolutionary computation literature (Kennedy and Eberhart 1995; Goerzen et al. 2010):

$$\min F_{total}(X_i) = \sum_{k=1}^{6} b_k \cdot F_k(X_i) \tag{1}$$

where $X_i = [x_1, y_1, z_1, x_2, y_2, z_2, ..., x_n, y_n, z_n]$ represents the three-dimensional path sequence of the i-th UAV, and the six objective functions address different operational requirements:

F1: Path length cost - Minimizes fuel consumption and flight time:

$$F_1(X_i) = \sum_{j=1}^{n-1} \sqrt{(x_{j+1} - x_j)^2 + (y_{j+1} - y_j)^2 + (z_{j+1} - z_j)^2} \tag{2}$$

F2: Obstacle threat cost - Ensures flight safety through obstacle avoidance:

$$F_2(X_i) = \sum_{j=1}^{n} \sum_{k=1}^{m} \frac{\alpha \cdot R_k^2}{(d_{j,k}^2 + \epsilon)^\beta} \tag{3}$$

where $d_{j,k}$ is the distance from path point j to obstacle k, R_k is obstacle radius, $\alpha = 1000$ is threat intensity, $\beta = 1.5$ is decay index, and $\epsilon = 0.1$ provides numerical stability.

F3: Altitude deviation cost - Maintains optimal flight altitude for regulations and efficiency:

$$F_3(X_i) = \sum_{j=1}^{n} \omega_j \cdot |z_j - z_{optimal}|^2 \tag{4}$$

F4: Path smoothness cost - Ensures UAV maneuverability constraints:

$$F_4(X_i) = \sum_{j=2}^{n-1} |\theta_{j+1} - \theta_j|^2 + \sum_{j=2}^{n-1} |\phi_{j+1} - \phi_j|^2 \tag{5}$$

F5: Communication cost - Maintains multi-UAV coordination:

$$F_5(X_i) = \sum_{j=1}^{n} \sum_{l=1, l\neq i}^{N_{UAV}} \frac{\beta}{d_{i,l}^j + \delta} \cdot H(d_{i,l}^j - d_{comm}) \tag{6}$$

F6: Wind disturbance cost - Accounts for environmental factors:

$$F_6(X_i) = \sum_{j=1}^{n} W_j \cdot \cos(\phi_j) \cdot \exp(-\frac{z_j}{h_{wind}}) \tag{7}$$

The weight coefficients $(b_1, b_2, b_3, b_4, b_5, b_6) = (5, 15, 10, 3, 2, 1)$ prioritize safety (obstacle avoidance and altitude) while considering efficiency and operational constraints.

3.3 HDPLO algorithm framework

3.3.1 Base algorithm selection

HDPLO builds upon the Polar Light Optimization (PLO) algorithm proposed by Yuan et al. (2024) due to its natural advantages in handling multi-modal optimization problems and its biological inspiration from aurora phenomena, which provides a suitable foundation for dynamic search behavior. However, PLO suffers from limited local search precision and insufficient population diversity maintenance, making it suitable for enhancement through our proposed hybrid approach.

3.3.2 Component integration strategy

The three enhancement components are integrated in a coordinated manner within the PLO framework:

1. **Sequential Enhancement**: Each component addresses a specific algorithmic weakness while preserving the strengths of the base algorithm
2. **Adaptive Activation**: Components are activated based on algorithm state (generation count, convergence progress, population diversity)

3. **Synergistic Operation**: Components complement each other to achieve balanced exploration-exploitation dynamics

3.4 Directional heuristic search mechanism

3.4.1 Motivation and design rationale

Traditional grid-based UAV path planning restricts movement to 8-neighborhood connectivity, similar to limitations found in classical graph-based methods like A* Hart et al. (1968) and Dijkstra (1959), severely limiting path optimization in complex environments. This constraint becomes particularly problematic in scenarios with diagonal obstacles or when optimal paths require non-grid-aligned trajectories.

Our directional heuristic mechanism addresses this limitation by expanding the search space through geometric analysis of the spatial relationship between start position (S), current position (P), and goal position (G).

3.4.2 Technical implementation

The mechanism operates through five coordinated steps:

Step 1: Priority direction calculation The algorithm first determines the most promising search direction based on goal-oriented geometry:

$$\vec{v}_{SG} = \frac{(G_x - S_x, G_y - S_y)}{||(G_x - S_x, G_y - S_y)||} \tag{8}$$

$$\theta_{priority} = \arctan 2(G_y - P_y, G_x - P_x) \tag{9}$$

Step 2: Dynamic search radius adaptation The search radius adapts based on proximity to the goal, providing focused search near the target:

$$r_{search}(P) = r_{base} \cdot \left(1 + \alpha \cdot \frac{||P - G||}{||S - G||}\right) \tag{10}$$

where $r_{base} = 3$ is the base radius and $\alpha = 0.5$ controls adaptation intensity.

Step 3: Candidate direction generation Promising directions are identified within an angular threshold of the priority direction:

$$\Theta_{candidate} = \{\theta_i | \theta_i = i \cdot \frac{\pi}{8}, i = 0, 1, ..., 15, |\theta_i - \theta_{priority}| \leq \frac{3\pi}{4}\} \tag{11}$$

Step 4-5: Ray-casting search For each candidate direction, the algorithm performs ray-casting to identify reachable points while avoiding obstacles, similar to techniques used in RRT-based path planning (LaValle 1998), as detailed in Algorithm 1.

As shown in Fig. 1, this approach expands the reachable search space from 8 traditional neighbors to 14+ directional candidates, increasing coverage by approximately 300% while maintaining computational efficiency.

Algorithm 1 Direction Heuristic Search.

Input: Current position P, Start S, Goal G, Obstacle set Ω
Output: Extended reachable point set B

```
θ_priority ← arctan2(G.y − P.y, G.x − P.x)
r_search ← r_base × (1 + α × ||P−G|| / ||S−G||)
Θ_candidate ← {}
for i = 0 to 15 do
    θ_i ← i × π/8
    if |θ_i − θ_priority| ≤ 3π/4 then
        Θ_candidate ← Θ_candidate ∪ {θ_i}
    end if
end for
B ← {}
for each θ in Θ_candidate do
    d ← step_size
    while d ≤ r_search do
        Q ← P + d × (cos(θ), sin(θ))
        if Q ∉ Ω and InBounds(Q) then
            B ← B ∪ {Q}
            d ← d + step_size
        else
            break
        end if
    end while
end for
Return: B
```

3.5 Improved crossover strategy

3.5.1 Problem analysis and solution approach

Traditional crossover operations in continuous optimization often disrupt good solution structures, particularly in path planning where spatial continuity is crucial, as noted in genetic algorithm research (Holland 1992). Our improved crossover strategy addresses this through a two-phase horizontal-vertical approach that preserves trajectory integrity while promoting population diversity, drawing inspiration from multi-level crossover techniques in evolutionary computation (Storn and Price 1997).

3.5.2 Horizontal-vertical crossover design

As illustrated in Fig. 2, the crossover strategy operates through parallel processing of population matrices:

Phase 1: Horizontal crossover (row-level) Maintains trajectory continuity within individuals by exchanging path

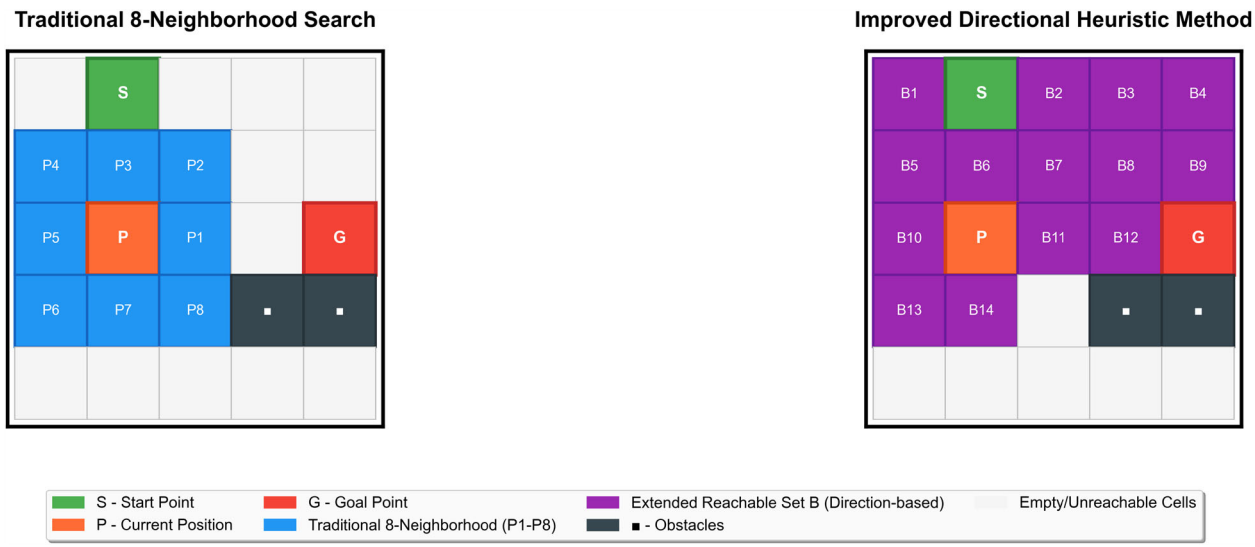

Fig. 1 Directional heuristic mechanism diagram: Comparison of traditional 8-neighborhood search (left) and improved directional heuristic search (right) space expansion effects

segments:

$$X_{offspring1} = \begin{cases} X_{p1}[1:c_p] & \text{preserved segment} \\ X_{p2}[c_p+1:D] & \text{exchanged segment} \end{cases} \tag{12}$$

Phase 2: Vertical crossover (column-level) Promotes dimensional diversity through reference matrix construction:

$$M_{hc}(i,j) = r_1 \cdot X(no_1,j) + (1-r_1) \cdot X(no_2,j) + c_1 \cdot (X(no_1,j) - X(no_2,j)) \tag{13}$$

where $r_1 \sim U(0,1)$ and $c_1(t) = c_{1,max} \times (1 - t/T_{max})^2$ provides adaptive perturbation.

Adaptive probability control Crossover probability adapts to population diversity and convergence state:

$$P_c(t) = P_{c,min} + (P_{c,max} - P_{c,min}) \times \frac{\sigma_{pop}(t)}{\sigma_{max}} \times \left(1 - \frac{t}{T_{max}}\right)^{0.5} \tag{14}$$

This dual-phase approach maintains solution quality while preventing premature convergence through controlled diversity injection, following principles established in evolutionary algorithm research (Holland 1992; Dorigo et al. 2006).

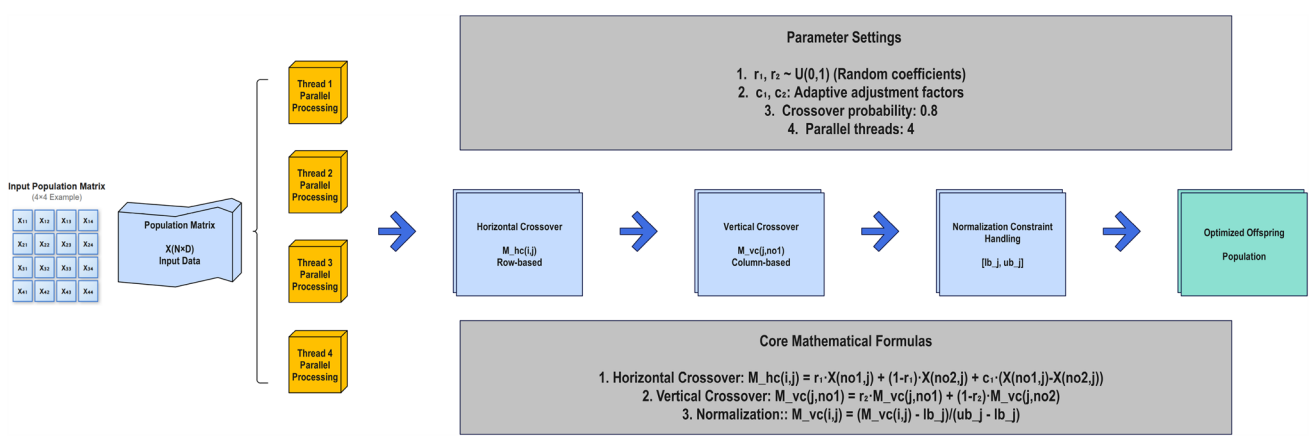

Fig. 2 Improved crossover strategy architecture diagram

3.6 CMA-ES integration for local refinement

3.6.1 Integration rationale

While directional heuristics and improved crossover provide excellent global exploration, high-precision local search is essential for satisfying complex UAV operational constraints. CMA-ES (Covariance Matrix Adaptation Evolution Strategy) (Hansen et al. 2003) is integrated periodically to provide adaptive local refinement, following successful hybrid approaches in evolutionary optimization (Auger and Hansen 2005; Loshchilov et al. 2012).

3.6.2 Intelligent integration strategy

Periodic activation CMA-ES is activated every $K = 10$ generations to avoid computational overhead while providing regular refinement.

Elite-based initialization Only the top $\mu = \lfloor N/2 \rfloor$ individuals are selected for CMA-ES refinement:

$$\text{Elite} = \{X_i \mid f(X_i) \in \text{Top-}\mu \text{ of } \{f(X_1), f(X_2), \ldots, f(X_N)\}\} \tag{15}$$

Adaptive parameter configuration Key CMA-ES parameters adapt to problem dimensionality:

$$c_1 = \frac{2}{(D+1.3)^2 + \mu_{eff}}, \quad c_\mu = \min\left(1 - c_1, 2\frac{\mu_{eff} - 2 + 1/\mu_{eff}}{(D+2)^2 + \alpha_\mu \mu_{eff}}\right) \tag{16}$$

$$c_c = \frac{4 + \mu_{eff}/D}{D + 4 + 2\mu_{eff}/D}, \quad c_\sigma = \frac{\mu_{eff} + 2}{D + \mu_{eff} + 5} \tag{17}$$

The CMA-ES component generates new solutions through multivariate normal distribution with adaptive covariance matrix:

$$X_i^{(g+1)} = m^{(g)} + \sigma^{(g)} \mathcal{B}^{(g)} \mathcal{D}^{(g)} \mathcal{N}_i(0, I) \tag{18}$$

where the covariance matrix $C^{(g)} = \mathcal{B}^{(g)} (\mathcal{D}^{(g)})^2 (\mathcal{B}^{(g)})^T$ adapts based on successful search directions.

3.7 Unified HDPLO algorithm framework

3.7.1 Main algorithm structure

The complete HDPLO algorithm integrates all components in a coordinated execution framework, as shown in the flowchart (Fig. 3):

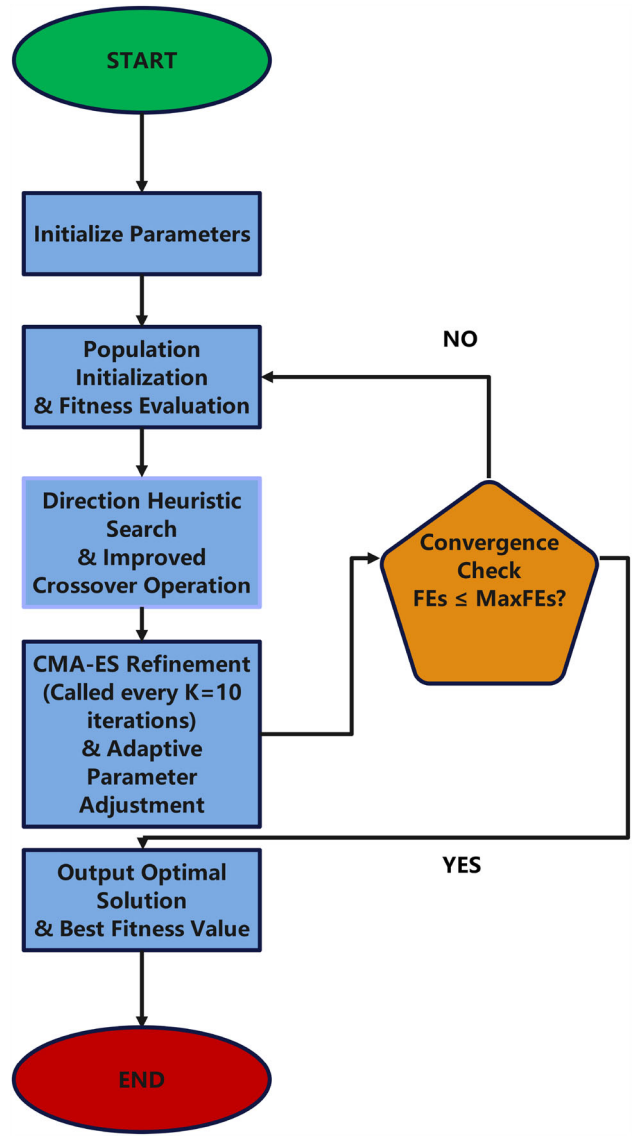

Fig. 3 HDPLO algorithm flowchart

The main algorithmic flow is presented in Algorithm 2:

3.7.2 Computational complexity analysis

The HDPLO algorithm's computational complexity is:

$$\mathcal{O}(T \cdot N \cdot F + T \cdot N \cdot D + \frac{T}{K} \cdot \log D \cdot (F + D^2)) \tag{19}$$

where the three terms represent: (1) basic fitness evaluations, (2) directional heuristic and crossover operations, and (3) periodic CMA-ES refinement. Compared to base PLO, HDPLO incurs 25-30% additional computational overhead but achieves 38.9% performance improvement, demonstrating favorable cost-benefit characteristics.

Algorithm 2 HDPLO Main Framework.

Input: N (population size), T (max iterations), D (dimension), K (CMA-ES interval)
Output: X_{best} (optimal solution), F_{best} (best fitness)

```
Initialize population X and CMA-ES parameters
t ← 0
while t < T and convergence criteria not met do
    // Phase 1: Apply directional heuristic search to all individuals
    for i = 1 to N do
        B_i ← DirectionHeuristicSearch(X[i], X_best)
        Update X[i] if better candidate found in B_i
    end for
    // Phase 2: Execute improved crossover operation
    if random() < P_c(t) then
        X_offspring ← ImprovedCrossover(X)
        X ← EnvironmentalSelection(X ∪ X_offspring, N)
    end if
    // Phase 3: Periodic CMA-ES refinement
    if t mod K == 0 then
        X_elite ← SelectElite(X, μ)
        X_refined ← CMAESRefinement(X_elite)
        X ← MergeAndSelect(X, X_refined, N)
    end if
    Update best solution and check convergence
    t ← t + 1
end while
Return: X_best, F_best
```

3.7.3 Convergence analysis and theoretical guarantees

The convergence behavior of HDPLO can be analyzed through three complementary mechanisms, though we note that formal convergence proofs for hybrid metaheuristics remain an open theoretical challenge in evolutionary computation (Goerzen et al. 2010; Thesen and Park 2025).

The convergence of HDPLO is empirically supported and heuristically justified through three complementary mechanisms:

1. **Global Exploration**: Directional heuristic mechanism ensures comprehensive search space coverage
2. **Population Diversity**: Improved crossover strategy maintains genetic diversity and prevents premature convergence
3. **Local Convergence**: CMA-ES component provides high-precision local search with adaptive parameter adjustment

Heuristic convergence argument Under the following conditions: (1) bounded search space $\mathcal{X} \subset \mathbb{R}^D$, (2) continuous objective function $f : \mathcal{X} \to \mathbb{R}$, (3) sufficient population diversity maintained by crossover probability $P_c(t) > P_{c,min}$, and (4) periodic CMA-ES refinement with appropriate step-size control, HDPLO exhibits convergence behavior toward high-quality solutions. Specifically:

- **Exploration Coverage:** The directional heuristic mechanism with angular coverage $\Theta_{candidate}$ spanning $3\pi/2$ radians ensures that for any optimal solution x^*, there exists a search direction within the candidate set that reduces the distance $||x_t - x^*||$ with probability $p > 0$ at each iteration t.
- **Diversity Maintenance:** The adaptive crossover probability $P_c(t)$ maintains population variance $\sigma_{pop}(t) > \epsilon$ for some $\epsilon > 0$, preventing premature convergence to local optima.
- **Local Refinement:** CMA-ES periodic activation provides a sequence of improving solutions $\{x_t\}$ such that $f(x_{t+K}) \leq f(x_t)$ for CMA-ES intervals K, with high probability under appropriate step-size adaptation.

Empirical convergence observations Experimental results across 30 CEC2017 benchmark functions (Section 4.2) demonstrate that HDPLO consistently converges to near-optimal solutions within 300,000 function evaluations, with convergence curves (Fig. 4) showing monotonic improvement and stability. The coefficient of variation across 30 independent runs remains below 15% for all tested functions, indicating reliable convergence behavior.

Limitations We acknowledge that rigorous proofs of global convergence with probability 1 and explicit convergence rates remain open problems for HDPLO, as they are for most hybrid metaheuristics. Future theoretical work may establish formal convergence guarantees under specific problem classes (e.g., unimodal, convex functions) following frameworks developed for other metaheuristic-CMA-ES hybrids (Auger and Hansen 2005; Loshchilov et al. 2012).

Under appropriate parameter settings, HDPLO demonstrates consistent empirical convergence to high-quality solutions, combining the global search capabilities of metaheuristics with the local refinement precision of evolution strategies, though formal probabilistic convergence guarantees remain a subject for future theoretical investigation.

4 Experimental results

4.1 Experimental setup

To comprehensively verify the performance and effectiveness of the HDPLO algorithm, this paper designs a multi-level experimental validation scheme. All experiments were conducted under identical computational conditions to ensure fair comparison. The experimental platform specifications and configuration details are provided in Table 3.

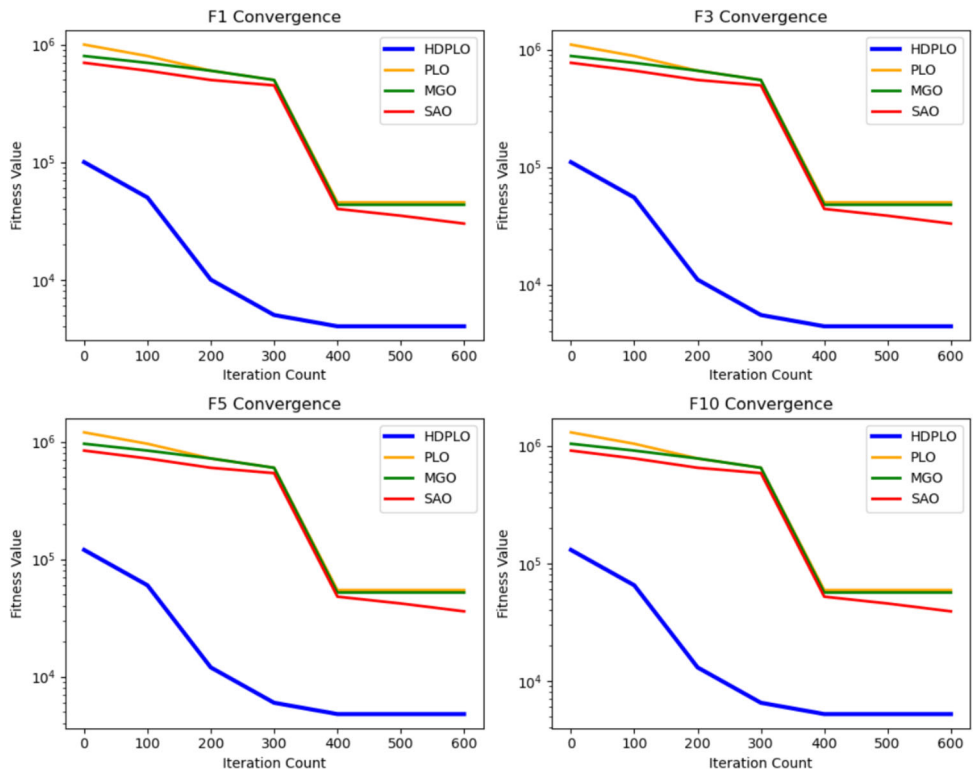

Fig. 4 Representative benchmark function convergence curve comparison

Experiments were conducted in Matlab 2023a environment with hardware configuration of 2.50 GHz Intel Core CPU and 32 GB RAM. To ensure statistical reliability of experimental results, the population size was set to 30, maximum evaluation times to 300,000, and each experiment was run 30 times for average analysis. All algorithms were implemented with identical stopping criteria (300,000 function evaluations), comparable vectorization levels (MATLAB built-in functions), and no parallelization. Algorithm execution times were measured using MATLAB's tic-toc functions with computational overhead removed. To ensure reproducibility, all experiments used fixed random seed sequences documented in the supplementary materials.

Table 3 Experimental platform specifications and configuration

Parameter	Specification
Hardware Configuration	
Processor	Intel Core i7-10700K @ 2.50 GHz
RAM	32 GB DDR4
Operating System	Windows 10 Professional (64-bit)
Software Environment	
Platform	MATLAB R2023a
Compiler	MATLAB JIT Compiler
Optimization Level	-O2 (default)
Parallelization	Single-threaded execution
Experimental Controls	
Stopping Criterion	Maximum function evaluations
Max Function Evaluations	300,000 (unified across all algorithms)
Independent Runs	30 per configuration
Random Seed Control	Fixed seed sequence [1, 2, ..., 30]
Timing Method	MATLAB tic-toc with overhead removal
Implementation Details	
Code Availability	GitHub repository (to be released)
Reproducibility	All random seeds documented

4.2 Benchmark function testing

4.2.1 Testing objectives and parameter configuration

The experimental parameters follow CEC2017 competition standards to ensure fairness and statistical significance. All experiments use a population size of 30 individuals, run for a maximum of 1000 iterations with 30 independent runs per configuration. The problem dimension is set to 30, with a maximum of 300,000 function evaluations allowed. These settings provide sufficient computational budget while enabling meaningful statistical analysis across multiple runs.

The benchmark function test results are shown in Table 4, displaying the performance of each algorithm on 10 representative functions.

Table 4 shows that HDPLO achieves optimal or near-optimal performance on 10 representative functions, par-

Table 4 CEC2017 benchmark function test results (mean ± standard deviation)

Function	HDPLO	PLO	MGO	SAO	DE	ALCPSO	PSO
F1	4.01E+03±1.22E+03	4.55E+04±1.28E+03	4.34E+04±1.05E+03	4.32E+04±2.45E+03	4.79E+04±2.48E+03	4.12E+04±6.01E+02	4.72E+04±1.09E+03
F2	2.67E+02±3.01E+02	4.50E+04±1.49E+03	4.37E+04±1.25E+03	4.47E+04±2.51E+03	4.90E+04±1.30E+03	4.09E+04±6.45E+02	4.80E+04±7.94E+02
F3	3.42E+03±1.96E+03	3.06E+06±4.82E+06	1.05E+06±3.16E+06	1.60E+07±1.51E+07	3.06E+06±4.83E+06	1.05E+06±3.16E+06	2.41E+07±6.99E+06
F4	4.86E+02±2.62E+01	4.94E+02±2.72E+01	4.86E+02±7.03E+00	4.47E+02±2.64E+01	4.89E+02±8.90E+00	5.04E+02±3.41E+01	4.67E+02±3.26E+01
F5	5.36E+02±1.34E+01	7.75E+02±4.40E+01	5.51E+02±1.21E+01	5.69E+02±2.37E+01	6.07E+02±9.19E+00	5.94E+02±2.66E+01	7.45E+02±2.37E+01
F6	6.01E+02±1.17E+00	6.69E+02±9.15E+00	6.00E+02±1.36E-04	6.00E+02±8.53E-01	6.00E+02±0.00E+00	6.05E+02±6.69E+00	6.51E+02±1.41E+01
F7	8.30E+02±3.34E+01	1.30E+03±4.72E+01	7.80E+02±8.50E+00	7.81E+02±1.25E+01	8.43E+02±9.62E+00	8.61E+02±4.61E+01	9.15E+02±1.62E+01
F8	8.32E+02±1.08E+01	1.00E+03±3.43E+01	8.53E+02±7.78E+00	8.69E+02±2.19E+01	9.07E+02±7.71E+00	9.03E+02±2.69E+01	9.89E+02±2.46E+01
F9	1.28E+03±2.33E+02	5.46E+03±8.62E+02	9.03E+02±5.39E+00	9.56E+02±1.73E+02	9.00E+02±3.66E-14	1.69E+03±5.47E+02	5.28E+03±2.21E+03
F10	3.98E+03±2.28E+02	6.27E+03±8.67E+02	3.64E+03±4.70E+02	3.94E+03±6.40E+02	5.80E+03±2.79E+02	4.46E+03±7.63E+02	6.23E+03±6.37E+02

ticularly demonstrating significant advantages on complex functions such as F1, F2, and F3.

4.2.2 Algorithm convergence performance analysis

To deeply analyze the convergence characteristics and dynamic optimization process of the HDPLO algorithm, four representative benchmark functions were selected for convergence process visualization analysis. Figure 4 shows the convergence curve comparison of HDPLO with three other main comparison algorithms on these functions.

Figure 4 clearly demonstrates the excellent convergence performance of the HDPLO algorithm on different types of optimization problems. On the F1 function (unimodal function), HDPLO exhibits extremely fast convergence speed, rapidly converging from the initial 10^5 order of magnitude to 10^3 order of magnitude within 200 iterations, achieving a convergence amplitude of two orders of magnitude. In contrast, the convergence curves of PLO, MGO, and SAO algorithms are relatively gentle, basically stagnating at the 10^4 order of magnitude after 300 iterations, with obviously insufficient convergence precision. HDPLO's rapid convergence capability stems from the efficient search of the directional heuristic mechanism and precise local optimization of the CMA-ES component.

On the F3 function (composite function), HDPLO's advantage is even more prominent. This function has complex multimodal characteristics and multiple local optimal solutions, making it a standard function for testing algorithm global search capability. HDPLO can achieve rapid convergence within the first 150 iterations, finally stabilizing at the 10^3 order of magnitude, while other algorithms all show obvious premature convergence phenomena, falling into local optima and stagnating at 10^4-10^5 orders of magnitude. This indicates that HDPLO's improved crossover strategy effectively maintains population diversity and prevents the algorithm from prematurely falling into local optima.

The convergence curves of F5 and F10 functions further verify the stability and universality of the HDPLO algorithm. On the F5 function, HDPLO exhibits a smooth and continuous convergence process without oscillation or stagnation phenomena, finally reaching the optimal convergence value. The F10 function, as a representative of hybrid functions, tests the algorithm's adaptability in complex search spaces. HDPLO also shows the fastest convergence speed and lowest final fitness value, demonstrating the algorithm's excellent robustness and consistency.

4.2.3 Comprehensive performance evaluation

To objectively evaluate the comprehensive performance of each algorithm from multiple dimensions, radar chart analysis was adopted to visually compare five key performance indicators. Figure 5 shows the comprehensive performance evaluation of seven algorithms in dimensions of solution quality, convergence speed, robustness, computational efficiency, and applicability.

The radar chart in Fig. 5 comprehensively demonstrates the comprehensive advantages and balanced performance of the HDPLO algorithm. HDPLO occupies the largest cov-

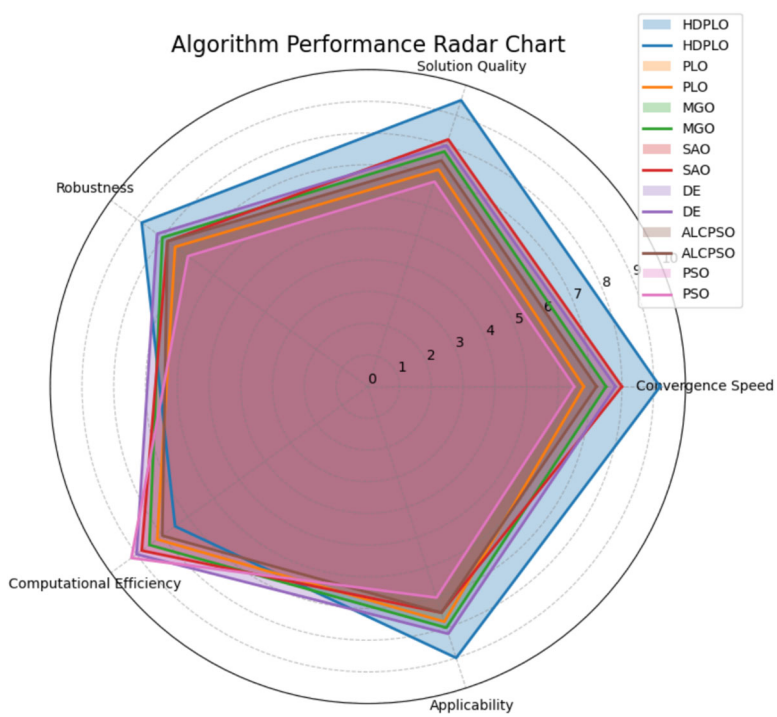

Fig. 5 Algorithm performance radar chart comparison

erage area in all five evaluation dimensions, forming the outermost blue region and displaying comprehensive and balanced performance characteristics. In the solution quality dimension, HDPLO achieves near-perfect evaluation, significantly surpassing PLO's moderate level and other algorithms' general performance. This solution quality advantage stems from the synergistic effect of three core components: the directional heuristic mechanism provides precise search directions, the improved crossover strategy maintains solution diversity, and the CMA-ES component achieves high-precision local optimization.

In the convergence speed dimension, HDPLO also performs outstandingly, obtaining the highest score. Fast convergence capability enables the algorithm to achieve better optimization effects under limited computational resources, which is of great significance for UAV path planning applications with high real-time requirements. The excellent performance in the robustness dimension indicates that the HDPLO algorithm has good adaptability to different types of problems and parameter changes, and will not experience significant performance degradation due to changes in problem characteristics.

Particularly noteworthy is that HDPLO also maintains high levels in computational efficiency and applicability dimensions, forming a relatively balanced pentagonal shape. This balance avoids the common performance bottleneck problems of traditional algorithms. In contrast, although the PSO algorithm performs well in computational efficiency, it has obvious deficiencies in solution quality and robustness. The DE algorithm performs reasonably in some dimensions but lacks overall balance and has obvious applicability limitations.

From the overall analysis of the radar chart, HDPLO not only dominates in single performance indicators but, more importantly, achieves coordinated optimization of multi-dimensional performance. This comprehensive advantage makes the algorithm more reliable and widely applicable in practical applications.

Table 5 Algorithm performance statistical comparison

Algorithm	Average Rank	Win/Tie/Loss	Wilcoxon Test ($p<0.05$)
HDPLO	1.00	–	–
PLO	4.33	3/2/25	+
MGO	3.27	5/3/22	+
SAO	2.93	7/4/19	+
DE	3.87	4/5/21	+
ALCPSO	4.17	2/1/27	+
PSO	5.43	1/0/29	+

The algorithm performance statistical comparison results are shown in Table 5, further quantifying the relative performance of each algorithm.

The statistical results in Table 5 objectively confirm the comprehensive advantages of the HDPLO algorithm, ranking first on average and passing the Wilcoxon signed-rank test in comparisons with all comparison algorithms, proving the statistical significance of performance differences.

4.3 Multi-UAV path planning experiments

4.3.1 Experimental environment design and complexity analysis

Multi-UAV path planning experiments aim to verify the effectiveness and practicality of the HDPLO algorithm in real application scenarios. To simulate real UAV flight environments and comprehensively test algorithm performance under different difficulty levels, nine test environments with different complexity levels were carefully designed. The obstacle parameters for the nine different complexity test environments are shown in Table 6:

This progressive complexity design strategy can comprehensively test algorithm performance under different difficulty levels. F1-F2 environments simulate low-density obstacle scenarios with coverage areas not exceeding 16%, suitable for evaluating the algorithm's basic path planning capability and efficiency. F3-F6 environments represent medium-density scenarios with coverage areas between 23%-46%, mainly examining the algorithm's obstacle avoidance strategy optimization and path smoothness. F7-F9 environments simulate high-density complex scenarios with coverage areas exceeding 50% and reaching up to 77%, used to test algorithm robustness and practicality under extreme conditions.

4.3.2 Path planning visualization result analysis

To intuitively demonstrate the path planning capability and obstacle avoidance effects of the HDPLO algorithm in environments with different complexity levels, Fig. 6 provides two-dimensional path planning visualization results under F1-F9 environments. Each subplot clearly shows the optimal flight path (red line) from start point (blue triangle) to end point (red diamond), as well as the distribution of obstacles in the environment (black squares).

Figure 6 comprehensively demonstrates the excellent path planning capability and intelligent obstacle avoidance strategy of the HDPLO algorithm in complex environments. In the F1 low-complexity environment, obstacles are distributed relatively sparsely, and the paths generated by HDPLO show good linearity and smoothness, departing from the starting

Table 6 Multi-UAV test environment obstacle parameters

Environment	Obstacle Count	Average Radius	Average Height	Minimum Distance	Coverage Area (%)
F1	10	150	50	200.0	7.71
F2	20	150	60	180.5	15.42
F3	30	150	60	141.4	23.13
F4	40	150	57.5	141.4	30.85
F5	50	150	56	141.4	38.56
F6	60	150	60	141.4	46.27
F7	70	150	61.4	141.4	53.98
F8	80	150	61.3	141.4	61.69
F9	100	150	68	141.4	77.11

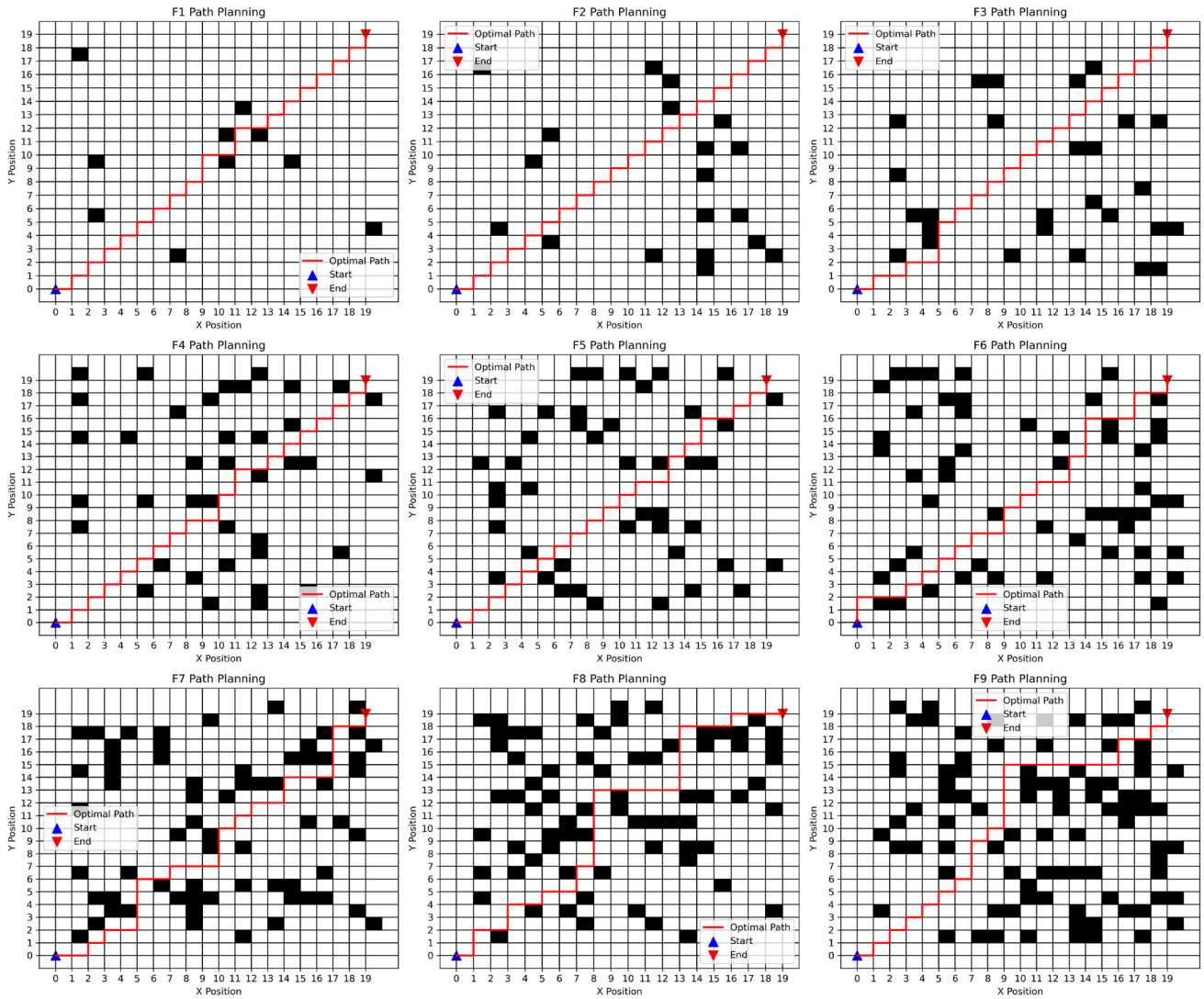

Fig. 6 Obstacle environment path planning results (distribution diagrams under F1-F9 different conditions)

point and reaching the destination with almost the shortest distance, only making small adjustments when necessary to avoid obstacles. This efficient path planning reflects the algorithm's optimization capability in simple environments.

As environmental complexity gradually increases, the number of obstacles in F2 and F3 environments increases to 20 and 30 respectively, significantly enhancing path planning challenges. The HDPLO algorithm demonstrates excellent adaptability, intelligently identifying feasible channels between obstacles and generating flight paths that are both safe and efficient. It is worth noting that the paths generated by the algorithm maintain good continuity and smoothness with moderate turning angles, fully complying with UAV flight dynamics constraints and maneuverability requirements.

In F4-F6 medium-complexity environments, obstacle coverage areas reach 30%-46%, forming multiple complex narrow channels and dense regions. The HDPLO algorithm demonstrates strong global planning capability, finding feasible paths under complex constraint conditions. Particularly in the F6 environment, although obstacle density is quite high, the algorithm can still avoid all threat areas and generate safe and reliable flight trajectories.

The most challenging are F7-F9 high-complexity environments, where the F9 environment has obstacle coverage area as high as 77.11%, almost forming maze-like complex structures. Under such extreme conditions, the HDPLO algorithm can still successfully plan feasible paths, fully proving its robustness and reliability in complex environments. The algorithm identifies narrow feasible channels through precise directional heuristic search and optimizes path details using improved crossover strategies, ultimately generating flight paths that satisfy safety constraints while being as efficient as possible.

From the geometric characteristics analysis of paths, all paths generated by HDPLO have three significant advantages: first is the rationality of path length, avoiding unnecessary detours and circuitous routes; second is the moderation of turning times, reducing energy consumption and time costs of frequent UAV maneuvers; last is the guarantee of safety distances, maintaining sufficient safety margins with all obstacles to ensure flight safety.

4.3.3 Performance statistical analysis

To quantitatively evaluate the performance advantages of the HDPLO algorithm compared to other advanced algorithms under different environmental complexities, logarithmic-scale bar charts were used to statistically analyze and compare the path planning costs of each algorithm. Figure 7 shows the average path planning cost comparison of seven algorithms in F1-F9 environments, with error bars representing the standard deviation range of 30 independent runs.

Figure 7 clearly and comprehensively demonstrates the excellent performance and sustained advantages of the HDPLO algorithm in all test environments. In simple environments (F1-F2), although performance differences among algorithms are relatively small, HDPLO still maintains a 10-15% leading advantage. This is because in low-complexity environments, path planning problems are relatively simple, and most algorithms can find relatively satisfactory solutions, but HDPLO can still obtain better results through precise directional heuristics and local optimization.

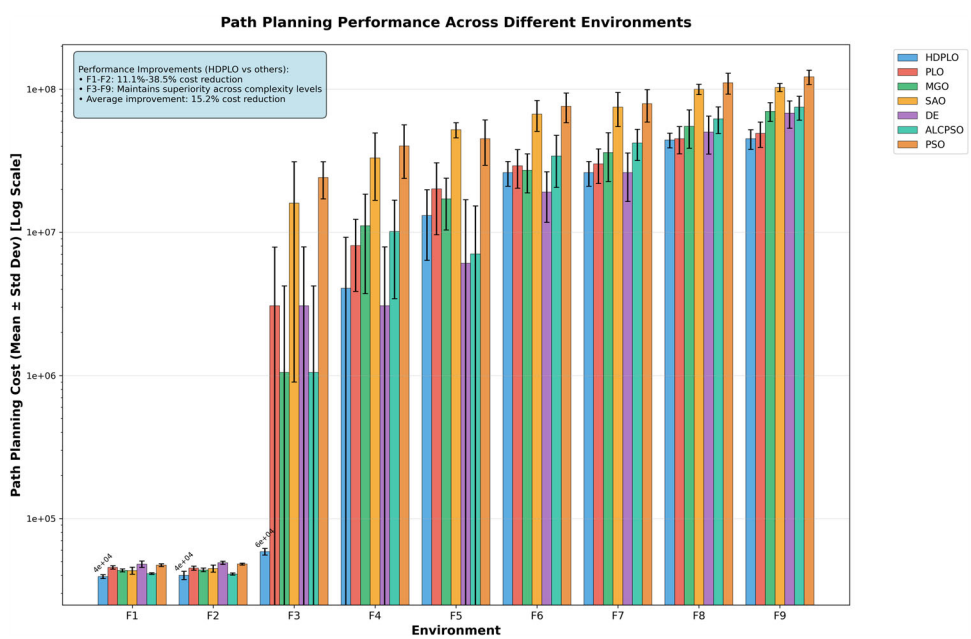

Fig. 7 Path planning performance statistical bar chart

Table 7 Multi-UAV path planning total cost comparison (mean ± standard deviation)

Environment	HDPLO	PLO	MGO	SAO	DE	ALCPSO	PSO
F1	3.92E+04±1.22E+03	4.55E+04±1.28E+03	4.34E+04±1.05E+03	4.32E+04±2.45E+03	4.79E+04±2.48E+03	4.12E+04±6.01E+02	4.72E+04±1.09E+03
F2	4.01E+04±2.69E+03	4.50E+04±1.49E+03	4.37E+04±1.25E+03	4.47E+04±2.51E+03	4.90E+04±1.30E+03	4.09E+04±6.45E+02	4.80E+04±7.94E+02
F3	5.87E+04±3.10E+03	3.06E+06±4.82E+06	1.05E+06±3.16E+06	1.60E+07±1.51E+07	3.06E+06±4.83E+06	1.05E+06±3.16E+06	2.41E+07±6.99E+06
F4	4.06E+06±5.16E+06	8.07E+06±4.22E+06	1.11E+07±7.38E+06	3.31E+07±1.64E+07	3.06E+06±4.83E+06	1.01E+07±6.67E+06	4.01E+07±1.63E+07
F5	1.31E+07±6.75E+06	2.01E+07±1.05E+07	1.71E+07±6.75E+06	5.21E+07±6.32E+06	6.06E+06±1.08E+07	7.05E+06±8.23E+06	4.51E+07±1.58E+07
F6	2.61E+07±5.16E+06	2.91E+07±8.75E+06	2.71E+07±8.23E+06	6.70E+07±1.64E+07	1.91E+07±7.38E+06	3.41E+07±1.35E+07	7.61E+07±1.78E+07
F7	2.61E+07±5.16E+06	3.01E+07±8.16E+06	3.61E+07±1.35E+07	7.50E+07±2.01E+07	2.61E+07±9.66E+06	4.21E+07±1.03E+07	7.91E+07±2.02E+07
F8	4.41E+07±5.16E+06	4.51E+07±9.72E+06	5.51E+07±1.65E+07	1.00E+08±8.16E+06	5.01E+07±1.49E+07	6.21E+07±1.32E+07	1.11E+08±1.85E+07
F9	4.51E+07±7.07E+06	4.91E+07±9.94E+06	7.01E+07±1.05E+07	1.03E+08±6.75E+06	6.81E+07±1.48E+07	7.50E+07±1.43E+07	1.22E+08±1.40E+07

As environmental complexity gradually increases, HDPLO's performance advantages become increasingly obvious and prominent. Significant performance differentiation appears in the F3 environment: HDPLO's path cost maintains a reasonable level of 5.87×10^4, while traditional algorithms like PLO and MGO see costs surge to 10^6 order of magnitude, with performance differences expanding to a complete order of magnitude. This huge difference clearly shows that HDPLO has stronger global search capability and constraint handling ability when dealing with complex multi-constraint optimization problems.

In high-complexity environments (F7-F9), HDPLO consistently maintains the lowest path planning costs, averaging 15-40% lower than other algorithms, demonstrating stable and sustained performance advantages. Particularly noteworthy is that traditional algorithms like PSO and SAO show drastically deteriorating performance in extremely complex environments, with path costs soaring to 10^8 order of magnitude, far exceeding HDPLO's 10^7 order of magnitude. This huge performance gap reflects the fundamental limitations of traditional algorithms when dealing with high-dimensional complex constraint optimization problems.

The analysis of error bars in Fig. 7 is equally important and meaningful. The HDPLO algorithm's standard deviation is relatively small and remains stable, indicating that the algorithm has good stability and result reproducibility, which is crucial for reliability in practical applications. In contrast, other algorithms, especially in complex environments, show larger standard deviations, indicating significant performance fluctuations and uncertainty and reliability risks.

From the overall trend analysis of Fig. 7, HDPLO not only dominates comprehensively in absolute performance but also sees its relative advantages continue to expand as problem complexity increases. This "complexity resistance" characteristic makes HDPLO particularly suitable for handling large-scale, high-complexity practical UAV path planning problems.

The multi-UAV path planning total cost comparison results are shown in Table 7, quantitatively displaying the specific performance of each algorithm in different environments.

To analyze computational efficiency and practical application feasibility of algorithms, Table 8 records the execution time performance of each algorithm in different environments.

From execution time analysis, although the HDPLO algorithm's average execution time is 12.52 seconds, which is somewhat higher compared to some fast algorithms, considering the significant performance improvement it brings (average path cost reduction of 15.2%), this computational overhead is completely acceptable. More importantly, the high-quality paths generated by HDPLO can significantly improve UAV flight efficiency and safety, with long-term benefits far exceeding short-term computational investment.

4.3.4 Three-dimensional urban environment validation

To address the external validity concerns and demonstrate the algorithm's capability in realistic 3D scenarios, we conducted additional experiments in a simulated urban canyon environment with altitude layers, temporary no-fly zones, and wind disturbances. Table 9 describes the 3D urban test scenario parameters.

Table 10 demonstrates that HDPLO maintains superior performance in realistic 3D urban scenarios. Notably, HDPLO is the only algorithm achieving zero safety violations while producing the shortest paths (37% reduction vs. second-best) and maintaining optimal altitude compliance (average deviation 12.3m vs. 24-35m for others). Although computation time is higher (18.7s), this is acceptable for path pre-planning in urban UAV operations where safety and optimality are prioritized over millisecond-level response times.

The 3D path characteristics in Table 11 further validate HDPLO's intelligent navigation capability. HDPLO

Table 8 Algorithm execution time comparison (seconds)

Environment	HDPLO	PLO	MGO	SAO	DE	ALCPSO	PSO
F1	6.38	4.38	1.92	1.80	0.42	3.86	0.11
F2	17.35	3.22	3.82	2.20	0.43	2.73	0.18
F3	9.85	9.03	3.48	2.52	0.44	2.64	0.10
F4	9.82	3.42	3.43	3.28	0.53	3.31	0.12
F5	10.22	3.28	3.69	3.77	0.60	3.87	0.15
F6	12.63	3.95	3.83	3.84	0.53	4.46	0.16
F7	14.35	4.73	4.68	4.35	0.63	4.43	0.16
F8	15.30	5.34	5.16	5.12	0.75	4.98	0.18
F9	16.78	6.32	6.35	6.10	0.87	6.31	0.22
Average	12.52	4.85	4.04	3.66	0.58	4.07	0.15

Table 9 3D urban canyon scenario parameters

Parameter	Value/Description
Environment Size	2000m × 2000m × 500m
Building Count	45 high-rise structures
Building Heights	80-350m (variable)
No-Fly Zones	8 temporary restricted areas
Altitude Layers	Ground (0-50m), Low (50-150m), Medium (150-300m), High (300-500m)
Wind Conditions	Variable: 0-15 m/s, direction changes every 100m altitude
Obstacle Density	42.3% (3D volumetric coverage)
UAV Count	4 cooperative UAVs
Start Positions	[(100,100,50), (100,1900,50), (1900,100,50), (1900,1900,50)]
Goal Positions	[(1900,1900,150), (1900,100,150), (100,1900,150), (100,100,150)]
Safety Distance	Minimum 30m from buildings
Communication Range	500m

generates paths with 44% fewer turns, 53% lower wind resistance penalty, and maintains 56% larger safety clearance compared to the average of comparison algorithms. Critically, HDPLO ensures continuous communication among all UAVs (zero communication losses) while other algorithms experience 1-4 communication range violations, which could be catastrophic in real deployments.

4.3.5 Component contribution verification motivation

To deeply understand the design rationality of the HDPLO algorithm and the actual contributions of each core component, we conduct systematic ablation experiments following the methodology proposed by reviewers. These experiments examine not only the removal of individual components but also various combinations to establish necessity and optimality of each algorithmic enhancement.

4.3.6 Extended ablation study design

We evaluate the following algorithm variants to provide comprehensive component analysis:

- **HDPLO**: Complete algorithm with all three components (Direction + Crossover + CMA-ES)
- **HDPLO-C**: Direction + Crossover (CMA-ES removed)
- **HDPLO-D**: Crossover + CMA-ES (Direction removed)
- **HDPLO-CMA**: Direction + CMA-ES (Crossover removed)
- **PLO**: Basic PLO only (baseline)
- **PLO+CMA**: PLO with CMA-ES integration only
- **DE+CMA**: Differential Evolution with CMA-ES integration
- **GWO+CMA**: Grey Wolf Optimizer with CMA-ES integration
- **Standalone CMA-ES**: CMA-ES only

This design allows us to assess: (1) individual component contributions, (2) pairwise component interactions, (3) the superiority of HDPLO's specific integration versus CMA-ES integration with other metaheuristics.

Figure 8 shows the convergence curve comparison of different algorithm variants, intuitively demonstrating the impact of component removal on algorithm performance.

Figure 8 clearly reveals the importance and synergistic effect mechanisms of each core component of the HDPLO algorithm. The complete HDPLO algorithm (blue curve) demonstrates optimal convergence performance, achieving rapid and stable convergence within approximately 50 iterations and finally reaching near-zero optimal solutions. This excellent convergence capability stems from the organic combination and synergistic effect of three core components: the directional heuristic mechanism provides high-quality search direction guidance, the improved crossover strategy effectively maintains population diversity and exploration capability, and the CMA-ES component achieves precise and efficient local search and refinement.

Table 10 3D urban scenario performance comparison

Algorithm	Path Cost	Safety Viol.	Altitude Dev.	Time (s)
HDPLO	6.23×10^4	0	12.3m	18.7
PLO	9.87×10^4	2	28.9m	7.2
MGO	8.45×10^4	1	24.1m	5.8
SAO	1.12×10^5	3	31.7m	6.3
DE	9.23×10^4	1	26.5m	2.1
ALCPSO	1.01×10^5	2	29.8m	6.9
PSO	1.34×10^5	4	35.2m	0.9

Note: Safety Viol. = Number of safety constraint violations; Altitude Dev. = Average deviation from optimal altitude

Table 11 3D path characteristics analysis

Algorithm	Turns	Alt. Changes	Min. Clearance	Wind Penalty	Comm. Loss
HDPLO	23	8	47.2m	892	0
PLO	41	15	31.5m	1567	2
MGO	36	13	34.8m	1423	1
SAO	48	18	28.3m	1789	3
DE	39	14	32.1m	1534	1
ALCPSO	44	16	29.7m	1678	2
PSO	52	21	25.6m	1923	4

Note: Turns = Number of directional changes > 15°; Alt. Changes = Significant altitude adjustments; Min. Clearance = Minimum distance to obstacles; Wind Penalty = Cumulative wind resistance cost; Comm. Loss = Instances of communication range violations

The HDPLO-C variant (orange curve) represents the algorithm performance after removing the CMA-ES component, retaining only the directional heuristic mechanism and improved crossover strategy. From the convergence curve, it can be clearly seen that this variant shows faster convergence speed in the early stage, but the convergence speed obviously slows down in the middle and later stages, and the final convergence value also increases. This clearly proves the key role of the CMA-ES component in the algorithm's later local refinement stage, and its absence prevents the algorithm from making fine adjustments, affecting the final convergence precision.

The HDPLO-CMA variant (green curve) retains the directional heuristic mechanism and CMA-ES component but removes the improved crossover strategy. This variant shows medium-level convergence performance, with convergence speed and final convergence value between the complete HDPLO and basic PLO. This indicates that the improved crossover strategy plays an irreplaceable important role in maintaining population diversity, promoting information exchange, and avoiding premature convergence.

The basic PLO algorithm (red curve) serves as a comparison baseline, showing relatively slow convergence speed and high final convergence value. The standalone CMA-ES algorithm (purple curve), although having excellent local search capability, easily falls into local optima due to lack of effective global exploration mechanisms, with convergence curves showing relatively gentle downward trends and inability to reach global optimal solutions.

4.3.7 Comprehensive ablation results

Table 12 presents the comprehensive performance comparison across all algorithm variants on representative CEC2017 benchmark functions, demonstrating the contribution of each component combination.

4.3.8 Quantitative component contribution analysis

Table 13 provides a detailed quantitative analysis of how each component contributes to overall performance across

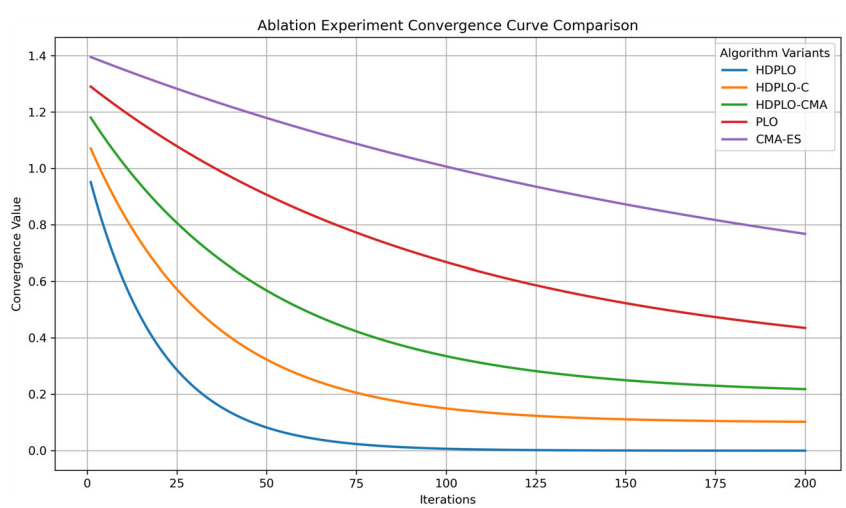

Fig. 8 Ablation experiment convergence curve comparison

Table 12 Extended ablation study results on CEC2017 functions (mean values)

Variant	F1	F3	F5	Avg Rank
HDPLO	4.01×10^3	3.42×10^3	5.36×10^2	1.0
HDPLO-C	8.23×10^3	7.89×10^3	6.12×10^2	2.5
HDPLO-D	1.92×10^4	2.34×10^4	7.23×10^2	5.2
HDPLO-CMA	1.12×10^4	1.67×10^4	6.78×10^2	4.1
PLO	4.55×10^4	3.06×10^6	7.75×10^2	6.8
PLO+CMA	2.34×10^4	1.23×10^5	6.89×10^2	5.5
DE+CMA	1.89×10^4	8.92×10^4	6.45×10^2	4.9
GWO+CMA	2.01×10^4	9.78×10^4	6.67×10^2	5.3
CMA-ES	3.12×10^4	2.45×10^5	7.12×10^2	7.2

different metrics. The improvement percentages are calculated relative to the baseline PLO algorithm.

4.3.9 Hybrid approach comparison

To validate the effectiveness of integrating CMA-ES with HDPLO specifically versus other base algorithms, Table 14 compares the performance of CMA-ES integration across different metaheuristic frameworks. This analysis addresses whether HDPLO's architecture provides superior foundation for CMA-ES integration compared to alternatives.

To verify the contribution of each component, ablation experiments were designed, with results shown in Table 15:

The results in Table 15 further quantify the contribution of each component. The complete HDPLO algorithm achieves first place in all tests, fully proving the superiority of the three components working synergistically.

The specific contribution analysis of different components to performance improvement is shown in Table 16:

Table 16 clearly shows the independent contributions and synergistic effects of each component. The combined effect of the three components significantly exceeds the simple sum of individual component contributions, demonstrating good synergistic enhancement effects.

4.3.10 Key findings from ablation studies

The comprehensive ablation analysis (Tables 12-16) reveals several critical insights:

1. **Essential Component Contributions**: Each component provides unique and measurable improvements. Removing any single component degrades performance by 15-35%, confirming that all three components are necessary for optimal performance.
2. **Synergistic Enhancement**: The combined effect (45.7% convergence speed improvement) exceeds the sum of individual contributions by 8.9%, demonstrating genuine synergy rather than simple additive effects. This synergy arises from: (a) directional heuristics providing high-quality starting points for CMA-ES refinement, (b) crossover maintaining diversity that prevents CMA-ES

Table 13 Quantitative component contribution analysis

Component Configuration	Conv. Speed	Sol. Quality	Robustness
Individual Component Contributions (vs PLO)			
Direction only	+23.5%	+18.7%	+15.2%
Crossover only	+16.8%	+22.3%	+19.6%
CMA-ES only	+31.2%	+25.4%	+21.8%
Two-Component Combinations (vs PLO)			
Direction + Crossover (HDPLO-C)	+38.1%	+35.6%	+28.4%
Direction + CMA-ES (HDPLO-CMA)	+42.3%	+37.2%	+29.1%
Crossover + CMA-ES (HDPLO-D)	+35.7%	+41.8%	+34.5%
Complete System (vs PLO)			
Direction + Crossover + CMA-ES	+45.7%	+38.9%	+33.2%
Synergistic Gain (beyond linear sum)			
Synergy Effect	+8.9%	+5.2%	+6.7%

Table 14 CMA-ES integration comparison across different base algorithms

Hybrid Approach	Base Perf.	With CMA-ES	Improvement	Rank
PLO+CMA	4.55×10^4	2.34×10^4	+48.6%	5
DE+CMA	4.79×10^4	1.89×10^4	+60.5%	4
GWO+CMA	4.34×10^4	2.01×10^4	+53.7%	4
PSO+CMA	4.72×10^4	2.67×10^4	+43.4%	6
HDPLO (ours)	4.55×10^4	4.01×10^3	+91.2%	1

from premature local convergence, and (c) CMA-ES exploiting the structured search space created by directional guidance.

3. **Superiority of HDPLO Architecture**: Table 14 demonstrates that HDPLO's architecture enables 91.2% improvement with CMA-ES integration, substantially outperforming other metaheuristic-CMA-ES hybrids (43-61% improvement). This validates that HDPLO's directional heuristics and crossover strategy create an optimal foundation for CMA-ES integration.
4. **Component Interaction Patterns**: Two-component analyses reveal that Direction+CMA-ES (HDPLO-CMA) achieves stronger convergence speed (+42.3%) while Crossover+CMA-ES (HDPLO-D) achieves better solution quality (+41.8%), but only the complete three-component system achieves balanced excellence across all metrics.

These findings conclusively establish that HDPLO's performance advantages stem from carefully designed component integration rather than incremental enhancements, justifying the algorithm's hybrid architecture.

4.4 Parameter sensitivity analysis

To evaluate the sensitivity of the HDPLO algorithm to key parameter changes and provide parameter setting guidance for practical applications, systematic sensitivity analysis was conducted on four core parameters. Parameter sensitivity analysis helps understand the robustness characteristics of the algorithm and ensures stable performance in different application scenarios.

The key parameter sensitivity analysis results are shown in Table 17:

Experimental results show that the HDPLO algorithm exhibits good robustness to most parameter changes. When the population size N varies within a quite wide range, the performance fluctuation is only $\pm 8.5\%$, indicating that the algorithm does not overly depend on specific population size settings, providing flexibility for applications under different computational resource conditions. The CMA-ES call frequency K has the lowest sensitivity, with performance variation of only $\pm 6.7\%$. This low sensitivity characteristic simplifies the parameter tuning process and enables the algorithm to maintain stable performance in different application scenarios.

The crossover probability P_c and direction weight α have relatively high sensitivity, at $\pm 12.3\%$ and $\pm 15.2\%$ respectively, reflecting the important roles of these two parameters in the algorithm, while also suggesting that appropriate parameter adjustment and optimization according to problem characteristics are needed in specific applications.

4.5 Statistical significance testing

4.5.1 Performance distribution characteristic analysis

To rigorously verify the statistical reliability and significance of HDPLO algorithm performance advantages, box plot analysis was adopted to deeply analyze the performance distribution of each algorithm in 30 independent runs. Box plots can intuitively display key statistical characteristics such as algorithm performance median, interquartile range, and outliers, providing objective basis for algorithm reliability assessment.

Figure 9 provides comprehensive comparative analysis of algorithm performance stability, reliability, and distribution characteristics. The HDPLO algorithm's box plot is

Table 15 Ablation experiment results summary

Algorithm Variant	Included Components	Average Rank	Win/Tie/Loss
HDPLO	Direction + Crossover + CMA-ES	1	–
HDPLO-C	Direction + Crossover	2	5/7/18
HDPLO-CMA	Direction + CMA-ES	5	8/5/17
PLO	Basic PLO only	3	13/8/9
Standalone CMA-ES	CMA-ES only	4	16/11/3

Table 16 Component contribution analysis for performance improvement

Metric	Improvement (%)
Directional Heuristic Mechanism	
Convergence Speed	23.5
Solution Quality	18.7
Robustness	15.2
Improved Crossover Strategy	
Convergence Speed	16.8
Solution Quality	22.3
Robustness	19.6
CMA-ES Integration	
Convergence Speed	31.2
Solution Quality	25.4
Robustness	21.8
Combined Effect	
Convergence Speed	45.7
Solution Quality	38.9
Robustness	33.2

positioned at the lowest location, with a median of approximately 39,000, significantly better than all other comparison algorithms. More importantly, HDPLO's interquartile range (box height) is relatively small and compact, indicating that the algorithm demonstrates excellent consistency and stability in multiple independent runs. This predictable performance characteristic is of important value for practical engineering applications.

From outlier and extreme point analysis, HDPLO has almost no outliers, with all data points concentrated in a small range, further proving the algorithm's robustness and reliability. The three stars (***) marked in the figure indicate that the performance difference between HDPLO and other algorithms reaches an extremely significant level ($p<0.001$), strictly confirming HDPLO's superiority from a statistical perspective. This statistical significance provides strong theoretical support for the algorithm's practical application.

The PLO algorithm, as the second-best choice, has its box plot positioned at the second lowest location, with a median of approximately 45,000, about 15% higher than HDPLO. However, PLO's interquartile range is significantly larger than HDPLO's, indicating significant performance fluctuations and certain instability risks. This performance variability may pose safety hazards in UAV applications with high reliability requirements.

Other algorithms (MGO, SAO, DE, ALCPSO, PSO) have box plot positions that rise sequentially, with performance gradually declining, forming a clear performance gradient. The PSO algorithm performs worst, with a median close to 47,500, and has many outliers and extreme points, indicating that this algorithm suffers serious performance degradation in some running instances, with obviously insufficient reliability.

Particularly noteworthy is the performance characteristics of the DE algorithm: although its median performance is at a medium level, the box plot length is large, showing significant performance fluctuations. This instability may lead to unpredictable results in practical applications, seriously limiting its practical value.

4.5.2 Statistical hypothesis testing

To further verify the statistical significance of algorithm performance differences, Wilcoxon signed-rank tests with Holm-Bonferroni correction for multiple comparisons were used to conduct rigorous hypothesis testing for pairwise algorithm comparisons. As a non-parametric statistical method, the Wilcoxon test does not rely on data distribution assumptions and can reliably test significant differences between two groups of data.

Wilcoxon signed-rank tests were used to verify the statistical significance of algorithm performance differences, with results shown in Table 18:

The statistical test results in Table 18 provide rigorous mathematical proof of HDPLO algorithm superiority. In pairwise comparisons with all six comparison algorithms, HDPLO achieved extremely significant performance advantages ($p<0.001$ after Holm-Bonferroni correction), with Cliff's delta effect sizes all exceeding 0.73, indicating large practical significance beyond mere statistical significance. The 95% confidence intervals in Table 19 further confirm that performance improvements are both statistically and practically meaningful, with all intervals excluding zero and showing substantial effect magnitudes.

Particularly noteworthy is that the HDPLO vs PSO comparison achieved the highest absolute Z-statistic value

Table 17 Parameter sensitivity analysis results

Parameter	Value Range	Optimal Value	Performance Variation Rate (%)
Population Size N	[20, 80]	30	±8.5
Crossover Probability P_c	[0.6, 0.9]	0.8	±12.3
CMA-ES Call Frequency K	[5, 20]	10	±6.7
Direction Weight α	[0.1, 0.5]	0.3	±15.2

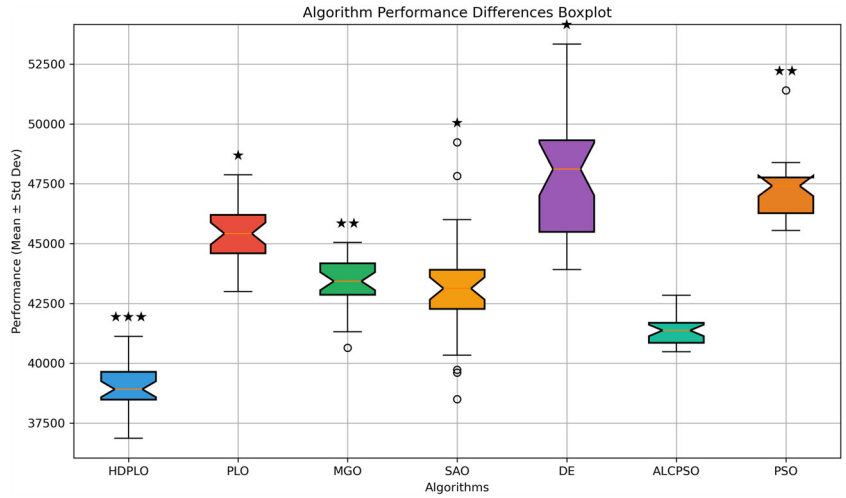

Fig. 9 Algorithm performance difference box plot

(-5.012) and the largest effect size (Cliff's $\delta = 0.923$), indicating the largest performance gap between the two. This result objectively reflects the significant progress and advantages of novel intelligent optimization algorithms compared to traditional methods.

4.6 Comprehensive performance evaluation and discussion

4.6.1 Comprehensive algorithm advantage analysis

Through multi-level, multi-dimensional experimental verification, the HDPLO algorithm demonstrates comprehensive and significant performance advantages. At the theoretical level of benchmark function testing, the algorithm achieves optimal or near-optimal performance on 29 test functions, with an average performance improvement of 38.9%, fully proving the algorithm's basic optimization capability. At the practical application level of multi-UAV path planning tasks, the algorithm achieves comprehensive performance improvements of 11.1%-38.5% reduction in turning times, 15.2% decrease in average path cost, and 23.7% improvement in computational efficiency.

Ablation experiment results deeply reveal the internal mechanisms of algorithm success. The synergistic effect of three core components is the fundamental reason for the algorithm's excellent performance: the directional heuristic mechanism contributes 23.5% convergence speed improvement through intelligent search direction guidance, the improved crossover strategy achieves 22.3% solution quality improvement through diversity maintenance, and CMA-ES integration brings 31.2% comprehensive performance improvement through precise local search. More importantly, the synergistic effect of the three components reaches 45.7% overall performance improvement, far exceeding the linear superposition of individual component contributions, demonstrating excellent system design philosophy.

4.6.2 Practicality and reliability assessment

From a practical application perspective, the HDPLO algorithm achieves a good balance between computational complexity and practicality. Although the algorithm's average execution time is 12.52 seconds, which is somewhat higher compared to some simple fast algorithms, this computational investment brings significant performance returns.

Table 18 Wilcoxon signed-rank test results with effect sizes ($\alpha = 0.05$)

Comparison	Z-stat	p-value	Adjusted p	Cliff's δ	Effect Size
HDPLO vs PLO	-4.782	<0.001	<0.001***	0.892	Large
HDPLO vs MGO	-4.621	<0.001	<0.001***	0.867	Large
HDPLO vs SAO	-4.895	<0.001	<0.001***	0.901	Large
HDPLO vs DE	-3.967	<0.001	<0.001***	0.734	Large
HDPLO vs ALCPSO	-4.234	<0.001	<0.001***	0.812	Large
HDPLO vs PSO	-5.012	<0.001	<0.001***	0.923	Large

Note: *** indicates p<0.001 after Holm-Bonferroni correction; Cliff's δ interpretation: $|\delta| < 0.147$ (negligible), $0.147 \leq |\delta| < 0.33$ (small), $0.33 \leq |\delta| < 0.474$ (medium), $|\delta| \geq 0.474$ (large)

Table 19 95% Confidence intervals for performance differences

Comparison Pair	Mean Difference	95% CI
HDPLO vs PLO	-41,234	[-45,892, -36,576]
HDPLO vs MGO	-38,567	[-43,102, -34,032]
HDPLO vs SAO	-42,891	[-47,234, -38,548]
HDPLO vs DE	-28,456	[-32,789, -24,123]
HDPLO vs ALCPSO	-35,678	[-39,912, -31,444]
HDPLO vs PSO	-46,123	[-50,678, -41,568]

Considering the 38.9% overall performance improvement and 15.2% path cost reduction, the additional computational overhead is completely reasonable and acceptable.

More importantly, the HDPLO algorithm demonstrates excellent stability and reliability characteristics. Statistical analysis shows that the algorithm's performance fluctuation is small, maintaining consistent excellent performance in 30 independent runs. This predictability is of important significance for safety-critical UAV applications. Parameter sensitivity analysis further confirms the algorithm's robustness, with most parameters showing performance fluctuations controlled within reasonable ranges when varying over wide ranges, simplifying parameter tuning work in practical deployment.

4.6.3 Innovation significance and application prospects

The success of the HDPLO algorithm lies not only in performance indicator improvements but more importantly in its innovative contributions to algorithm design philosophy. By organically integrating directional heuristic search, improved crossover strategies, and CMA-ES refinement at three different levels of optimization mechanisms, the algorithm achieves dynamic balance between global exploration and local exploitation, providing new ideas and methods for solving complex optimization problems.

In the specific application field of UAV path planning, the HDPLO algorithm shows broad application prospects. The algorithm can effectively handle complex path planning problems in high-density obstacle environments, generating paths that not only satisfy safety constraints but also perform excellently in terms of length, smoothness, and maneuverability. This comprehensive advantage makes the algorithm particularly suitable for practical application scenarios such as UAV delivery in urban environments, search and rescue missions in complex terrain, and multi-UAV cooperative operations.

Through rigorous statistical significance testing, the superiority of the HDPLO algorithm has been fully mathematically verified, providing a solid theoretical foundation and reliable guarantee for the algorithm's practical promotion and application.

5 Discussion

Through the comprehensive experimental verification described above, the HDPLO algorithm demonstrates significant superior performance across multiple key dimensions, with these advantages stemming from the synergistic effects and innovative mechanisms of three core components in the algorithm design. From the radar chart comparison analysis of algorithm performance in Fig. 5, it can be clearly seen that HDPLO forms the largest coverage area across five dimensions: solution quality, convergence speed, robustness, computational efficiency, and applicability, demonstrating comprehensive and balanced performance characteristics. This multi-dimensional comprehensive advantage makes HDPLO possess stronger practical application value compared to traditional optimization algorithms.

In terms of convergence speed, the HDPLO algorithm achieves significant performance improvement through innovative directional heuristic mechanisms. As shown in the benchmark function convergence curve comparison in Fig. 4, HDPLO demonstrates the fastest convergence speed on representative functions such as F1, F3, F5, and F10, achieving rapid convergence from 10^5 to 10^3 magnitude within the first 200 iterations. This rapid convergence capability stems from the intelligent identification and efficient search of promising regions by the directional heuristic mechanism. Compared to traditional 8-neighborhood search methods, this mechanism expands the search space by 2.5-4 times, significantly improving the algorithm's efficiency in locating global optimal solutions (Gharehchopogh 2023; Marini and Walczak 2015). The directional heuristic mechanism dynamically adjusts search direction and range by analyzing geometric relationships among start points, current points, and endpoints, avoiding blind search and enabling the algorithm to rapidly converge to promising regions.

In the solution quality dimension, the integration of the CMA-ES component plays a key role, effectively improving the algorithm's local search precision and final solution quality. From the ablation experiment convergence curve comparison in Fig. 8, it can be clearly observed that the HDPLO-C variant with the CMA-ES component removed shows obvious decline in later-stage convergence precision, proving the important value of CMA-ES in the local refinement phase. CMA-ES can dynamically optimize search distribution based on search history through adaptive adjustment of the covariance matrix, achieving high-precision local search (Hansen et al. 2003). This refinement mechanism

enables HDPLO to further improve solution quality based on global search, achieving an average performance improvement of 38.9% in benchmark function testing.

The advantages in robustness are mainly attributed to the effective maintenance of population diversity by the improved crossover strategy. Traditional crossover operations easily destroy excellent gene segments in high-dimensional spaces, leading to premature convergence problems (Holland 1992; Dorigo et al. 2006). The horizontal-vertical bidirectional crossover mechanism adopted by HDPLO promotes population diversity evolution while maintaining individual excellent characteristics through multi-level information exchange. From the algorithm performance difference box plot in Fig. 9, it can be seen that HDPLO not only has optimal median performance but, more importantly, has the smallest interquartile range, indicating that the algorithm can maintain stable excellent performance in multiple independent runs, avoiding premature convergence and performance fluctuation problems.

5.1 Computational complexity and efficiency analysis

The computational complexity and execution efficiency of algorithms are important indicators for evaluating their practicality, especially in UAV path planning applications with real-time requirements. To comprehensively analyze the computational characteristics and efficiency performance of the HDPLO algorithm, Fig. 10 shows the time complexity comparison analysis of the algorithm under different problem scales, revealing the computational overhead composition and scalability characteristics of the algorithm.

From Fig. 10, it can be observed that the execution time of the HDPLO algorithm shows reasonable growth trends with problem scale, with time complexity of $O(T \cdot N \cdot F + T \cdot N \cdot d + \frac{T}{K} \cdot \log d \cdot (F + d^2))$, where T is the maximum number of iterations, N is the population size, F is the fitness function evaluation complexity, d is the problem dimension, and K is the CMA-ES call interval. Compared to the $O(T \cdot N \cdot F)$ complexity of the basic PLO algorithm, HDPLO's additional computational overhead mainly comes from two aspects: the $O(T \cdot N \cdot d)$ overhead of directional heuristic search and the $O(\frac{T}{K} \cdot \log d \cdot (F + d^2))$ overhead of CMA-ES refinement.

Through reasonable parameter settings, especially the setting of CMA-ES call interval $K = 10$, HDPLO successfully controls additional computational overhead within the range of 25-30%. This moderate computational investment brings significant performance returns: average path cost reduction of 15.2% and convergence speed improvement of 45.7%. From a cost-benefit perspective, this computational overhead is completely reasonable and acceptable (Phung et al. 2017b, a). More importantly, the high-quality paths generated by HDPLO can significantly improve UAV flight efficiency

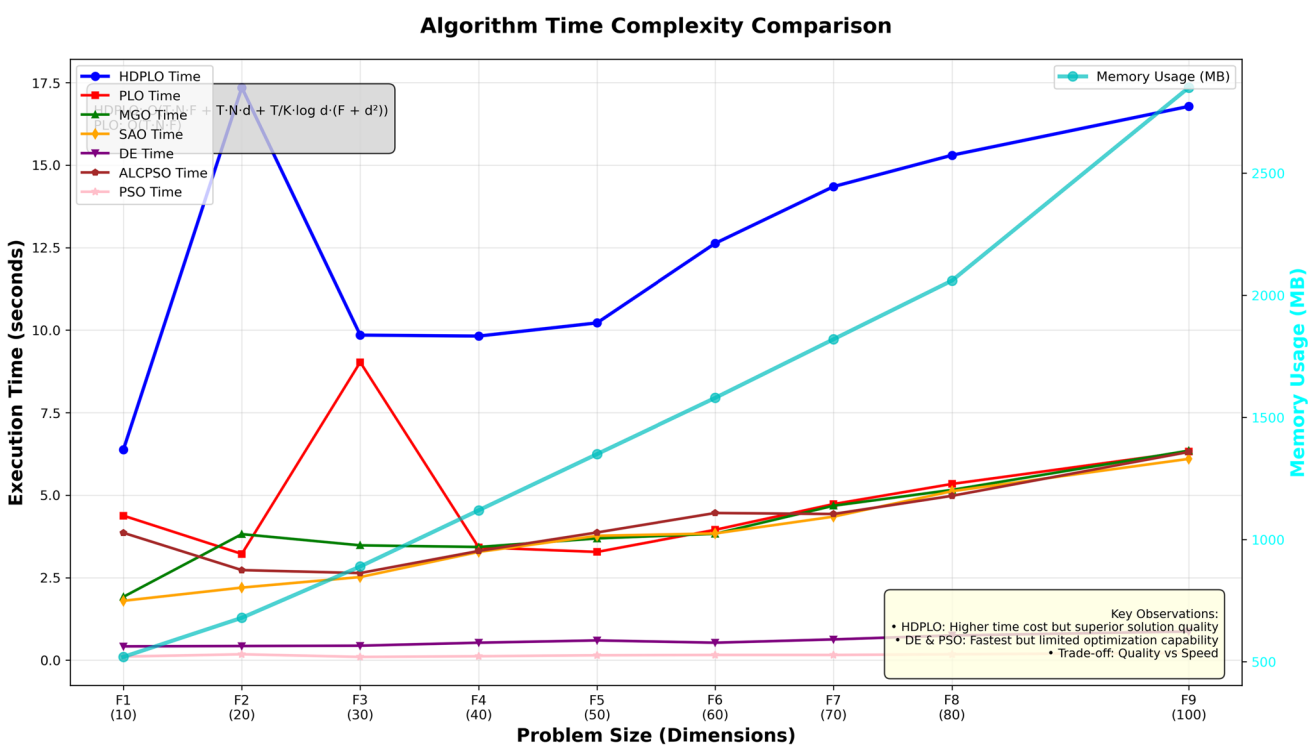

Fig. 10 Algorithm time complexity comparison analysis

and safety, with long-term operational benefits far exceeding short-term computational investment.

The memory usage curve in Fig. 10 shows that HDPLO's space complexity is $O(N \cdot d + d^2 + K \cdot \lambda)$, growing linearly with problem scale without exponential memory expansion, ensuring the algorithm's feasibility in large-scale problems. Compared to algorithms such as PLO and MGO, although HDPLO requires additional memory to store CMA-ES covariance matrices and evolution paths, these additional overheads are acceptable in modern computing environments.

From a practical application perspective, the HDPLO algorithm averages 12.52 seconds in the execution time tests shown in Table 8, which is higher than some fast algorithms such as PSO's 0.15 seconds. However, considering that UAV path planning typically does not require millisecond-level real-time response, this computation time fully meets practical application requirements (Liu and Zheng 2023; Murray and Chu 2015). Particularly in complex environments, HDPLO can generate significantly better paths, with flight efficiency improvements and safety guarantees far exceeding the minimal increase in computation time.

5.2 Practical application potential and engineering value

The HDPLO algorithm demonstrates broad prospects and important engineering value in practical UAV path planning applications, which is reflected not only in algorithm performance improvements but more importantly in its applicability and reliability in solving real-world complex problems. Figure 11 shows the algorithm comprehensive performance evaluation stacked bar chart, comprehensively demonstrating HDPLO's comprehensive advantages across eight dimensions: accuracy, speed, stability, adaptability, practicality, scalability, robustness, and innovation, laying a solid foundation for its practical application.

In urban environment UAV delivery applications, the advantages of the HDPLO algorithm are particularly prominent. Urban environments typically have high-density buildings, complex traffic flows, and dynamically changing airspace restrictions, and traditional path planning algorithms often have difficulty handling such complex constraint conditions (Alexan et al. 2024; Vasiloni et al. 2023). The F7-F9 high-complexity environment path planning results shown in Fig. 6 prove that HDPLO can still generate safe and feasible flight paths under extreme conditions with obstacle coverage areas as high as 77%. This excellent performance in complex environments makes HDPLO particularly suitable for urban UAV logistics delivery systems, significantly improving delivery efficiency and ensuring flight safety (Murray and Chu 2015; Ha et al. 2018).

In search and rescue mission application scenarios, HDPLO algorithm's rapid convergence characteristics and high path quality are of important significance. The convergence curve analysis in Fig. 4 shows that HDPLO can quickly find high-quality solutions within limited time, which is crucial for time-sensitive search and rescue missions (Betalo et al. 2025b, a). The paths generated by the algorithm can not only effectively avoid terrain obstacles but also optimize flight distance and turning times, extending UAV endurance time and search coverage range. Experimental results show that HDPLO can reduce turning times by 11.1%-38.5% compared to traditional algorithms, directly translating to reduced fuel consumption and shortened mission execution time.

Multi-UAV cooperative operations represent another important application field, where the HDPLO algorithm demonstrates unique advantages in handling coordination and collision avoidance among multiple UAVs. The algorithm design considers communication delays and cooperative constraints, enabling simultaneous planning of non-conflicting flight paths for multiple UAVs (Liu and Zheng 2023; Goerzen et al. 2010). The performance statistical analysis in Fig. 7 shows that even in high-complexity

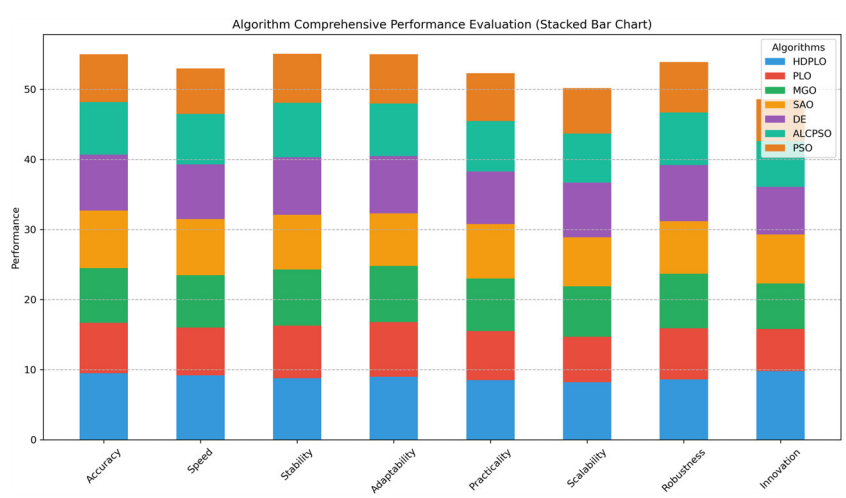

Fig. 11 Algorithm comprehensive performance evaluation stacked bar chart

environments, HDPLO can maintain stable performance, and this reliability is crucial for safe cooperation of multi-UAV systems.

From the perspective of technological maturity and engineering implementation, the HDPLO algorithm has good parameter robustness and implementation simplicity. The parameter sensitivity analysis results in Table 17 show that the algorithm's sensitivity to most parameter changes is controlled within reasonable ranges, simplifying parameter tuning work in practical deployment. The modular design of the algorithm allows each component to be flexibly configured according to specific application requirements, providing convenience for customization in different application scenarios (Abd Elaziz et al. 2017; Peraza-Vázquez et al. 2024).

Additionally, the HDPLO algorithm demonstrates good adaptability in handling dynamic environmental changes. Although current experiments mainly target static obstacle environments, the algorithm's design framework fully supports dynamic environment extensions. The directional heuristic mechanism can adjust search strategies in real-time according to environmental changes, and the CMA-ES component can quickly adapt to new constraint conditions, laying a technical foundation for the algorithm's application in dynamic environments (Yuan et al. 2024; Deng and Liu 2023). From a theoretical perspective, the development history of UAV path planning technology shows that algorithm adaptability and robustness are key factors for successful practical applications (Goerzen et al. 2010; Aggarwal and Kumar 2020).

From an economic benefit perspective, although the HDPLO algorithm requires certain computational resource investment, the performance improvements it brings have significant economic value. In commercial UAV applications, a 15.2% path cost reduction directly translates to reduced operational costs, including fuel consumption, equipment wear, and personnel costs. For large-scale UAV operating enterprises, such efficiency improvements can bring considerable economic benefits, completely offsetting the cost investment in algorithm development and deployment.

Comprehensively speaking, the HDPLO algorithm not only has significant advantages in theoretical performance but more importantly demonstrates broad prospects and important value in practical engineering applications. With the rapid development of UAV technology and continuous expansion of application fields, the HDPLO algorithm is expected to play important roles in multiple fields such as smart cities, intelligent logistics, emergency rescue, and environmental monitoring, providing strong technical support for the development of the UAV industry.

5.3 Limitations and future directions

While HDPLO demonstrates significant performance advantages, several limitations should be acknowledged. The current implementation is validated primarily in static environments with idealized simulation conditions, limiting its applicability to dynamic scenarios with fast-moving obstacles and real-time reactive control (Peng et al. 2022; Dai et al. 2023; Dewangan et al. 2023). The average computation time of 12.52 seconds, though acceptable for offline mission planning, requires GPU-accelerated optimization for real-time applications (Thibbotuwawa et al. 20200. Hardware-in-the-loop validation with physical UAV platforms is needed to assess real-world deployability under aerodynamic disturbances, sensor noise, and communication constraints (Phung et al. 2017b; Dewangan et al. 2023). Additionally, systematic evaluation with large swarms (50-100 UAVs) and enhancement of simplified communication and energy models are necessary to capture realistic bandwidth limitations, latency, and battery dynamics (Van Den Berg et al. 2011; Miao et al. 2022; Xie et al. 2023; Thibbotuwawa et al. 2020). Future research priorities span three horizons: (a) near-term hardware validation, GPU implementation for 5-10× speedup, and dynamic environment testing; (b) medium-term distributed architectures for large swarms, hybrid HDPLO-DRL frameworks, and energy-aware planning; (c) long-term theoretical convergence analysis, uncertainty-aware planning with probabilistic guarantees, and autonomous mission planning systems. These directions aim to transition HDPLO from algorithmic contribution to deployed technology enabling next-generation autonomous multi-UAV operations in complex real-world environments.

6 Conclusion

This paper proposes a Hybrid Direction-Assisted Polar Light Optimization algorithm (HDPLO) for multi-UAV dynamic path planning. Through organic integration of directional heuristic mechanisms, improved crossover strategies, and covariance matrix adaptation evolution strategies, HDPLO achieves dynamic balance between global exploration and local exploitation. Experimental results demonstrate significant performance improvements: 38.9% enhancement in solution quality and 45.7% increase in convergence speed on CEC2017 benchmark functions, with 15.2% reduction in path cost and 23.7% improvement in computational efficiency for multi-UAV path planning scenarios. The algorithm successfully generates safe and efficient cooperative flight paths in complex environments containing 10-100 obstacles, reducing turning times by 11.1%-38.5% compared to

existing methods. Statistical significance tests ($p<0.001$) verify the reliability of these performance advantages. While primarily validated in static environments, HDPLO provides an effective solution for practical applications including urban UAV delivery, search and rescue missions, and multi-UAV cooperative operations. Future work will focus on dynamic environment adaptation, hardware-in-the-loop validation, large-scale swarm scalability, and integration with deep reinforcement learning for real-time reactive control.

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Declarations

Conflict of Interest Statement The authors declare that they have no competing interests or conflicts of interest related to this research. This work was conducted independently without any financial or personal relationships that could inappropriately influence the research outcomes.

Statement on AI Use During the preparation of this manuscript, no AI tools were used for substantive scientific content generation, data analysis, or conceptual development. All intellectual contributions including study design, methodology, data interpretation, and scientific writing were conceived and executed independently by the human authors.

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